

Fundamental Anomalies*

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Abstract

This paper proposes a portfolio-independent method to estimate q -theory models, in which parameters are obtained using Bayesian Markov Chain Monte Carlo (MCMC) to match firm-level stock returns. Our methodology addresses [Campbell's \(2017\)](#) critique on prior studies that model parameters are chosen to fit a specific set of anomalies and different values are needed to fit each anomaly. By targeting the entire sample of firm-level returns and allowing industry and time variations in parameter values, our estimations yield higher correlations between realized and fundamental portfolio returns compared to prior literature. Additionally, the estimated two-capital model generates large and significant size, momentum, profitability, investment, and intangibles premiums, but falls short in explaining the value and accruals anomalies. This limitation underscores the importance of portfolio-independent parameter estimation in evaluating a model's capability to generate return anomalies.

Keywords: q -theory, Bayesian MCMC estimation, Anomalies, Investment, Profitability

1 Introduction

The investment-based asset pricing literature (Cochrane, 1991; Gomes, Kogan, and Zhang, 2003; Zhang, 2005, among others) studies returns from the supply side of the economy and formulates returns based on firm fundamentals, under the assumption that a firm operates to maximize its market equity.¹ Liu, Whited, and Zhang (2009) show that such a simple q -model fits value, earnings surprises, and investment anomalies well, when model parameters are chosen to match the average observed stock returns of decile portfolios sorted by these anomaly variables. Additional asset pricing anomalies and firm valuation ratios can be explained in this framework as shown in subsequent studies, such as Belo, Xue, and Zhang (2013), Liu and Zhang (2014), and Gonçalves, Xue, and Zhang (2020), among others.

In this line of research, model parameters are estimated via the Generalized Method of Moments (GMM) with the average observed returns of testing portfolios as target moments. This practice often generates parameter estimates that are portfolio-dependent. As Campbell (2017) (page 213) puts it: “This problem, that different parameters are needed to fit each anomaly, is a pervasive one in the q -theoretic asset pricing literature.” The set of parameters that are chosen to explain their target anomalies often fails to explain other anomalies. Therefore, this approach can overstate the model’s capability in explaining the target anomalies. A recent work by Gonçalves, Xue, and Zhang (2020) addresses Campbell’s critique by showing that a two-capital q -model estimated using a precise aggregation method fits well the value, momentum, profitability, and investment anomalies with a single set of structural parameters. In this study, we go a step further and propose a portfolio-independent estimation method.

To address Campbell (2017)’s critique, we propose a portfolio-independent methodology in which model parameters are chosen using the Bayesian Markov Chain Monte Carlo (MCMC) method to match the entire sample of individual firm-level stock returns. Bayesian MCMC is widely used in the field of macroeconomics (Smets and Wouters, 2007, among others) and has also been used to study return dynamics (Li, Wells, and Yu, 2008; Li, Li, Wang, and Yu, 2019, among others). Given its portfolio-independence, this methodology can be used to evaluate the

¹The investment-based asset pricing literature is voluminous and other examples include Berk, Green, and Naik (1999), Carlson, Fisher, and Giammarino (2004), Li, Livdan, and Zhang (2009), Papanikolaou (2011), Belo, Lin, and Bazdresch (2014), and Belo, Gala, Salomao, and Vitorino (2022), among others.

capability of a given model in explaining stock market anomalies.

For comparison with prior literature, we estimate the same two-capital q -model as in [Gonçalves, Xue, and Zhang \(2020\)](#) and examine its performance in explaining anomalies with portfolio-independent parameters. Using the 40 value, profitability, investment, and momentum decile portfolios as test assets, [Gonçalves, Xue, and Zhang \(2020\)](#) show that the estimated model can successfully generate these four anomaly premiums. Different from their approach, we estimate the same model to match firm-level stock returns instead and ask whether the estimated model can generate a variety of return anomalies in data. Our simulation studies show that Bayesian MCMC is able to discover the true parameter values under this model framework and performs better than the GMM and Nonlinear Least Squares (NLS) methods. With the flexibility of Bayesian MCMC, we estimate the model under four specifications, each with constant, industry-specific, time-varying, or industry-specific and time-varying parameter values.

To examine the capability of the model in explaining stock market anomalies, we consider 12 well-documented anomalies covering six major categories classified by [Hou, Xue, and Zhang \(2020\)](#):² value anomaly sorted on book-to-market equity ratio (BM); momentum anomaly sorted on the prior 11-month returns skipping the most recent month (R11); four investment anomalies sorted on asset growth (I/A), net stock issues (NSI), investment-to-assets ratio ($\Delta\text{PI}/A$), and accruals (Accruals); three profitability anomalies sorted on return-on-equity (ROE), return-on-assets (ROA), and gross profitability (GP/A); two intangibles anomalies sorted on R&D expenses-to-market ratio (RD/M) and advertising expenses-to-market ratio (Ad/M); and one trading frictions anomaly sorted on market capitalization (Size).

Several findings emerge from our study. First, the estimated q -model matches the mean, skewness, and kurtosis of firm-level returns well under all four specifications and is able to capture 30% to 56% of the volatility depending on the specification. The specification with

²We choose the 12 anomalies based on the following criteria: (1) in each of the six major categories, we select the anomaly variables that have been extensively documented to predict future returns and generate significant abnormal returns in our sample period (the only exception is size, which does not generate significant abnormal returns in our sample period but is kept due to its importance in the literature); (2) investment-based asset pricing models have been suggested by prior studies to explain these anomalies: value and size ([Gomes, Kogan, and Zhang, 2003](#); [Carlson, Fisher, and Giammarino, 2004](#); [Zhang, 2005](#)), momentum ([Liu and Zhang, 2014](#)), asset growth ([Watanabe et al., 2013](#); [Titman, Wei, and Xie, 2013](#)), investment-to-assets ratio and new stock issues ([Lyandres, Sun, and Zhang, 2008](#); [Li, Livdan, and Zhang, 2009](#)), accruals ([Wu, Zhang, and Zhang, 2010](#)), return-on-equity, return-on-assets, and gross profitability ([Kogan, Li, and Zhang, 2023](#); [Ai, Li, and Tong, 2021](#)), research and development expenses ([Li, 2011](#); [Lin, 2012](#)), and advertising expenses ([Belo, Lin, and Vitorino, 2014](#)).

industry-specific and time-varying parameter values leads to significantly lower mean absolute error (m.a.e.) than the other three specifications. We conduct an extensive set of tests to confirm that the industry and time variations in our estimates of the two model parameters, the investment adjustment cost parameter and production curvature parameter, are largely consistent with their economic interpretations.

Second, the model-implied firm-level stock returns (thereafter, fundamental returns) under all four specifications exhibit large and significant size, momentum, investment, profitability, and intangibles premiums.³ More importantly, the estimated models capture the dynamics of the 120 decile portfolios (10 for each anomaly) very well. Specifically, under the specification with industry-specific and time-varying parameter values, the returns of the fundamental and realized decile portfolio are all highly correlated (correlation coefficients are all significant at the 1% level). The average correlation coefficient is 0.69 for the decile portfolios and 0.43 for the high-minus-low portfolios. The comparison among the four specifications indicates that both the industry and time variations in parameter values are critical to the high correlations for the high-minus-low portfolios, while only time variation is critical to the high correlations for the decile portfolios.

These high correlations contrast sharply with prior studies, where the average anomaly premiums are well-matched but the correlations between fundamental and realized portfolio returns are generally close to zero, or even negative. This phenomenon is termed as “the correlation puzzle” by [Liu, Whited, and Zhang \(2009\)](#). Utilizing an advanced aggregation method, [Gonçalves, Xue, and Zhang \(2020\)](#) are able to generate mostly positive correlations. However, among the 44 BM, R11, I/A, and ROE decile and high-minus-low portfolios studied by [Gonçalves, Xue, and Zhang \(2020\)](#), only 10 out of 44 correlations are positive and significant (at least at the 10% level) as shown in Panel B of their Table 6, compared to 21 out of 44 under our specification with constant parameters. This comparison indicates that matching firm-level stock returns, rather than the average returns of test portfolios, also helps to generate higher correlations.

Last but not least, none of the four specifications generates significant value premium and

³Under the specification with industry-specific and time-varying parameters, the posterior means for each premium per annum and its t -statistic are as follows: the fundamental value premium is 0.46% ($t=0.26$), accruals premium is 4.74% ($t=4.45$), size premium is -5.99% ($t=-5.63$), the momentum premium is 11.82% ($t=12.51$), the I/A premium is -3.08% ($t=-2.25$), the NSI premium is -3.05% ($t=-3.36$), the Δ PI/A premium is -5.79% ($t=-4.81$), the ROE premium is 4.62% ($t=5.72$), the ROA premium is 3.80% ($t=3.99$), the GP/A premium is 7.26% ($t=5.84$), the RD/M premium is 5.24% ($t=2.12$), and the Ad/M premium is 7.46% ($t=2.82$). In particular, the sign of the fundamental accrual premium is the opposite of that in the data (-5.58% with $t=-3.14$).

accruals premium with the correct sign. The small estimated value of the investment adjustment costs parameter is the reason why our estimations fail to generate the value premium, whereas the one in [Gonçalves, Xue, and Zhang \(2020\)](#) successfully does. We use the same model and data sample as theirs, with the only difference being that their estimation targets the book-to-market decile portfolio returns, while we target firm-level returns. This result illustrates that a model’s ability to explain anomalies is likely overstated if its parameters are selected to match the anomaly returns it aims to explain. Depending on methods and datasets, estimates of the investment adjustment costs parameter in the literature cover a wide range. Our estimates are in similar magnitude as those in the influential work of [Cooper and Haltiwanger \(2006\)](#), among others. Note that our model’s failure to generate the value premium should not be interpreted as a failure of the investment-based asset pricing models because our model is highly stylized. Models in the asset pricing literature that generate the value premium often feature asymmetric investment adjustment costs, which is absent in our model.⁴

Our work is built directly on [Liu, Whited, and Zhang \(2009\)](#), [Liu and Zhang \(2014\)](#), and [Gonçalves, Xue, and Zhang \(2020\)](#). These papers conduct GMM estimations of various q -models using average anomaly portfolio returns as target moments. [Liu, Whited, and Zhang \(2009\)](#) show that a one-capital q -model can match the average returns of portfolios sorted on earnings surprises, book-to-market equity, and capital investment. [Liu and Zhang \(2014\)](#) use the same model and estimation procedure to explain the momentum premium. However, the estimated parameter values vary with testing portfolios substantially. [Gonçalves, Xue, and Zhang \(2020\)](#) estimate a two-capital q -model to match the average returns of 40 decile portfolios sorted on book-to-market equity, asset growth, return-on-equity, and momentum. They show that when fundamental returns are computed at firm level rather than at portfolio level, parameter estimates are more stable due to better aggregation. Different from these previous studies, our estimation method involves neither aggregation nor portfolios.

In a similar vein, [Belo, Xue, and Zhang \(2013\)](#) estimate a q -model by matching average q at the portfolio level, in addition to matching return moments. [Belo et al. \(2022\)](#) estimate a q -model with both tangible and intangible capitals by matching the time series of portfolio-level cross-sectional

⁴Table A.11 in the Internet Appendix lists 28 theoretical publications with costly investment reversibility in their models that have been published since 1999.

mean valuation ratios for a given set of testing portfolios. The estimation method in [Belo et al. \(2022\)](#) allows the dynamics of valuation ratios to be better captured. Our method can be easily applied to explain firm-level valuation ratios, which is a promising future direction.

Our paper is also related to the literature that questions the practice of evaluating asset pricing models using portfolios as testing assets. [Lo and MacKinlay \(1990\)](#) show that the particular way of portfolio grouping can dramatically influence the results of the asset pricing tests. [Lewellen, Nagel, and Shanken \(2010\)](#) point out that when testing portfolios have a strong factor structure, such as [Fama and French \(1992\)](#) 25 Size-BM portfolios, strong explanatory power for the average returns of these portfolios can provide weak support for a model. [Jegadeesha et al. \(2019\)](#) propose an instrumental variable approach to conduct empirical tests of asset pricing models using individual stocks. While they focus on tests of linear asset pricing models from the demand side, we directly estimate a model of firm-level returns from the supply side and our model naturally features time-varying risk premiums.

The rest of the paper is organized as follows. Section 2 outlines the model. Section 3 explains the data used in the estimation, describes the estimation procedure, and verifies the accuracy of Bayesian MCMC estimates under our model framework using simulation studies. Section 4 presents the estimation results and compares the performance of the four estimation specifications. Section 5 examines the performance of the model in terms of the anomaly premiums and the correlations between the fundamental and realized portfolio returns. Section 6 discusses the economic mechanisms behind the (in)capability of the estimated model in explaining anomalies. Section 7 concludes.

2 The Model

We adopt the two-capital model in [Gonçalves, Xue, and Zhang \(2020\)](#), in which firms use three inputs in production: long-term physical capital (K), short-term working capital (W), and costlessly adjustable input (S) such as energy and purchased service, the prices of which are taken as given by firms. Operating profit of firm i in industry j at time t is $\Pi_{it} = \Pi(K_{it}, W_{it}, S_{it})$, which exhibits constant-return-to-scale. Under the assumption of a perfectly competitive and frictionless market for input S , S_{it} is chosen to maximize contemporaneous operating profits. With

Cobb-Douglas production technology, marginal products of physical and working capital are given by $\partial \Pi_{it}/\partial K_{it} = \gamma_{jt}^K Y_{it}/K_{it}$ and $\partial \Pi_{it}/\partial W_{it} = \gamma_{jt}^W Y_{it}/W_{it}$, respectively, in which $\gamma_{jt}^K, \gamma_{jt}^W > 0$ are the corresponding shares of capital in sales Y_{it} with $\gamma_{jt} \equiv \gamma_{jt}^K + \gamma_{jt}^W < 1$.⁵ The model is estimated under four specifications, each with constant, industry-specific, time-varying, or industry-specific and time-varying parameter values. For generality, we formulate the model with industry-specific and time-varying parameter values in this section.

Note that the implications of the model hold regardless of whether model parameters are constant, industry-specific, or time-varying, as long as they are exogenous. We follow the convention in this line of research and assume rational expectation in the model, in which individuals know the true economic model, its parameters and shocks, and the nature of the stochastic processes that govern their evolution. The proofs and derivations in the Appendix do not rely on constant model parameters. The role of exogenous and time-varying parameters in the model is analogous to that of productivity shocks. The assumptions crucial to the model predictions are (1) profit function $\Pi(K_{it}, W_{it}, S_{it})$ is constant-return-to-scale; (2) investment adjustment cost function is linearly homogeneous in investment and capital. In the estimations with time-varying parameters, we assume that these parameters follow random walks.⁶

Firms choose investments in physical and working capital to maximize the market equity. Physical capital evolves as $K_{it+1} = (1 - \delta_{it})K_{it} + I_{it}$ in which I_{it} is the investment in physical capital, and δ_{it} is the depreciation rate. Investment in physical capital incurs quadratic adjustment costs:

$$\Phi_{it} \equiv \Phi(I_{it}, K_{it}) = \frac{a_{jt}}{2} \left(\frac{I_{it}}{K_{it}} \right)^2 K_{it}, \quad (1)$$

where a_{jt} is the physical adjustment costs parameter. Working capital evolves as $W_{it+1} = W_{it} + \Delta W_{it}$, in which ΔW_{it} is the investment in working capital. In addition, working capital does not depreciate and is not accompanied with adjustment costs.

In addition to equity financing, firm i in industry j issues debt B_{it+1} with interest rate r_{it+1}^B at the beginning of time t , which is repaid at the beginning of $t + 1$. At tax rate τ_t , firm i 's net payout is given by $D_{it} \equiv (1 - \tau_t)(\Pi_{it} - \Phi_{it}) - I_{it} - \Delta W_{it} + B_{it+1} - r_{it}^{Ba} B_{it} + \tau_t \delta_{it} K_{it}$, in which

⁵Section A in the Internet Appendix provides the proof.

⁶We also try a specification that parameters evolve AR(1) process. The estimation yields close to one persistences and for simplicity, we assume random walk for estimations with time-varying parameters.

$r_{it}^{Ba} \equiv r_{it}^B - \tau_t(r_{it}^B - 1)$ is the after-tax interest rate. Taking the stochastic pricing kernel, M_{t+1} , as given, firm i chooses I_{it} , K_{it+1} , ΔW_{it} , W_{it+1} , and B_{it+1} to maximize its cum-dividend market equity, $V_{it} \equiv E_t[\sum_{s=0}^{\infty} M_{t+s} D_{t+s}]$. The first-order condition for physical investment implies that $E_t[M_{t+1} r_{it+1}^K] = 1$, in which r_{it+1}^K is the return on physical capital investment:

$$r_{it+1}^K = \frac{(1 - \tau_{t+1}) \left[\gamma_{jt+1}^K \left(\frac{Y_{it+1}}{K_{it+1}} \right) + \frac{a_{jt+1}}{2} \left(\frac{I_{it+1}}{K_{it+1}} \right)^2 \right] + \tau_{t+1} \delta_{it+1} + (1 - \delta_{it+1}) \left[1 + (1 - \tau_{t+1}) a_{jt+1} \left(\frac{I_{it+1}}{K_{it+1}} \right) \right]}{1 + (1 - \tau_t) a_{jt} \left(\frac{I_{it}}{K_{it}} \right)}. \quad (2)$$

Similarly, the first-order condition for working capital investment implies that $E_t[M_{t+1} r_{it+1}^W] = 1$, in which r_{it+1}^W is the return on working capital investment:

$$r_{it+1}^W = 1 + (1 - \tau_{t+1}) \gamma_{jt+1}^W \frac{Y_{it+1}}{W_{it+1}}. \quad (3)$$

Section A in the Internet Appendix shows that the weighted average of the two investment returns equals the weighted average cost of equity and the after-tax cost of debt:

$$w_{it}^K r_{it+1}^K + (1 - w_{it}^K) r_{it+1}^W = w_{it}^B r_{it+1}^{Ba} + (1 - w_{it}^B) r_{it+1}^S, \quad (4)$$

in which $w_{it}^B \equiv B_{it+1}/(V_{it} - D_{it} + B_{it+1})$ is the firm's market leverage, $r_{it+1}^S \equiv V_{it+1}/(V_{it} - D_{it})$ is the stock return, $w_{it}^K \equiv q_{it}^K K_{it+1}/(q_{it}^K K_{it+1} + W_{it+1})$ is the weight of firm's market value attributed to physical capital and $q_{it}^K \equiv 1 + a_{jt}(1 - \tau_t) I_{it}/K_{it}$ is the marginal q of physical capital. The marginal q of working capital is one in the absence of adjustment costs in working capital investment. The Tobin's q of firm i at time t is the weighted average of marginal q 's of physical and working capitals, given by

$$q_{it} = \frac{P_{it} + B_{it+1}}{K_{it+1} + W_{it+1}} = \left[1 + a_{jt}(1 - \tau_t) \frac{I_{it}}{K_{it}} \right] \frac{K_{it+1}}{K_{it+1} + W_{it+1}} + \frac{W_{it+1}}{K_{it+1} + W_{it+1}}. \quad (5)$$

Solving for the stock return from equation (4) leads to the model-implied fundamental stock return

of firm i from t to $t + 1$:

$$\begin{aligned}
r_{it+1}^F &\equiv f(X_{it}, X_{it+1} | \theta_t, \theta_{t+1}) \\
&= \left\{ (1 - \tau_{t+1}) \left[\gamma_{jt+1} \left(\frac{Y_{it+1}}{K_{it+1}} \right) + \frac{a_{jt+1}}{2} \left(\frac{I_{it+1}}{K_{it+1}} \right)^2 \right] + \tau_{t+1} \delta_{it+1} \right. \\
&\quad \left. + (1 - \delta_{it+1}) \left[1 + (1 - \tau_{t+1}) a_{jt+1} \left(\frac{I_{it+1}}{K_{it+1}} \right) \right] \right. \\
&\quad \left. + \frac{W_{it+1}}{K_{it+1}} \right\} / \left\{ (1 - w_{it}^B) \left[1 + (1 - \tau_t) a_{jt} \left(\frac{I_{it}}{K_{it}} \right) + \frac{W_{it+1}}{K_{it+1}} \right] \right\} - \frac{w_{it}^B r_{it+1}^{Ba}}{1 - w_{it}^B}, \quad (6)
\end{aligned}$$

where X_{it} is the set of accounting variables used in equation (6) that represent firm i 's fundamentals, and $\theta_t \equiv \{(\gamma_{jt}, a_{jt}); j = 1, \dots, 10\}$ is the set of model parameters at time t for Fama-French 10 industries. The equality between the realized stock return and the model-implied fundamental return, $r_{it+1}^S = r_{it+1}^F$, holds for any firm i and for any period from t to $t + 1$ under this framework. Next, we estimate the two structural parameters, γ and a , based on this equality. Notice that γ^K and γ^W cannot be separately identified because r^F depends on their summation only.

3 Data and Estimation Methodology

In this section, we first explain how to construct the firm characteristics used in equation (6) and the 120 decile portfolios, 10 for each anomaly. We then introduce the Bayesian MCMC estimation and compare its performance with those of the GMM and NLS using simulation studies.

3.1 Data

Our sample includes all common stocks traded on NYSE, Amex, and NASDAQ with available accounting and return data. We exclude firms with primary standard industrial classifications between 6000 and 6999 (financial firms), firms with negative book equity, and firms with nonpositive total assets, net property, plant, and equipment, or sales at the portfolio formation. These data items are needed to calculate firm-level fundamental returns. We obtain monthly stock return data from the Center for Research in Security Prices (CRSP). Firm-level accounting data are obtained from the annual and quarterly Standard and Poor's Compustat industrial files. Our data sample covers the period from January 1967 to June 2017.

3.1.1 Anomalies We explore 12 anomalies covering all six categories defined in [Hou, Xue, and Zhang \(2020\)](#): value anomaly sorted on book-to-market equity ratio (BM); momentum anomaly sorted on the prior 11-month returns skipping the most recent month (R11); four investment anomalies sorted on asset growth (I/A), net stock issues (NSI), investment-to-assets ratio ($\Delta\text{PI}/A$), and accruals (Accruals); three profitability anomalies sorted on return-on-equity (ROE), return-on-assets (ROA), and gross profitability (GP/A); two intangibles anomalies sorted on R&D expense-to-market ratio (RD/M) and advertising expense-to-market ratio (Ad/M); and one trading frictions anomaly sorted on market capitalization (Size).⁷ We choose these 12 anomalies based on the following criteria: (1) The average value-weighted returns of their high-minus-low deciles with NYSE breakpoints are significant at the 5% level, with the exception of the size anomaly. Size anomaly is included since it is one of the most studied anomalies in the literature. (2) Investment-based asset pricing models have been suggested by prior studies to explain these anomalies, for example, value and size ([Gomes, Kogan, and Zhang, 2003](#); [Carlson, Fisher, and Giammarino, 2004](#); [Zhang, 2005](#)), momentum ([Liu and Zhang, 2014](#)), asset growth ([Watanabe et al., 2013](#); [Titman, Wei, and Xie, 2013](#)), investment-to-assets ratio and new stock issues ([Lyandres, Sun, and Zhang, 2008](#); [Li, Livdan, and Zhang, 2009](#)), accruals ([Wu, Zhang, and Zhang, 2010](#)), return-on-equity, return-on-assets, and gross profitability ([Kogan, Li, and Zhang, 2023](#); [Ai, Li, and Tong, 2021](#)), research and development expenses ([Li, 2011](#); [Lin, 2012](#)), and advertising expenses ([Belo, Lin, and Vitorino, 2014](#)). Appendix A provides the definitions of these variables and the construction of the corresponding decile portfolios. Table 1 presents the monthly average excess returns of the 10 decile portfolios sorted on each of the 12 anomaly variables. We confirm that all 12 anomalies except size have statistically and economically significant premiums in our sample period.

3.1.2 Measures and timing alignment Model-implied fundamental returns are constructed in annual frequency because the needed fundamental variables such as investments are only available at annual frequency for the long sample starting from 1967. In the model, time- t stock variables are at the beginning of year t , and time- t flow variables are over the course of year t . Thus, time- t stock variables are obtained from the balance sheet of fiscal year $t - 1$ and flow variables from the

⁷[Hou, Xue, and Zhang \(2020\)](#) show that almost all well-known anomalies in the trading frictions category cannot be successfully replicated. For completeness, we include the Size anomaly from this category since it is one of the most widely studied anomalies in the literature.

balance sheet of fiscal year t .

We adopt the same measures used by [Gonçalves, Xue, and Zhang \(2020\)](#) for the variables needed to construct the fundamental returns. Specifically, output, Y_{it} , is measured as sales (Compustat annual item SALE). Physical capital, K_{it} , is net property, plant, and equipment (item PPENT). Short-term working capital, W_{it} , is current assets (item ACT). Total debt, B_{it+1} , is long-term debt (item DLTT, zero if missing) plus short-term debt (item DLC, zero if missing) from fiscal year t balance sheet. Tax rate τ is the statutory corporate income tax rate from the commerce clearing house’s annual publications. The depreciation rate of physical capital, δ_{it} , is the amount of depreciation and amortization (item DP) minus the amortization of intangibles (item AM, zero if missing) divided by physical capital (item PPENT). Physical investment, I_{it} , is measured as $K_{it+1} - (1 - \delta_{it})K_{it}$. The market leverage, w_{it}^B , is the ratio of total debt to the sum of total debt and market equity. The pre-tax cost of debt, r_{it}^B , is the ratio of total interest and related expenses (item XINT) scaled by total debt, B_{it} . Following [Gonçalves, Xue, and Zhang \(2020\)](#), we winsorize unbounded variables, including I_{it}/K_{it} , I_{it+1}/K_{it+1} , $\Delta W_{it}/W_{it}$, $\Delta W_{it+1}/W_{it+1}$, at the 2.5% - 97.5% level. For variables bounded below by zero, including Y_{it+1}/K_{it+1} , Y_{it+1}/W_{it+1} , $Y_{it+1}/(K_{it+1} + W_{it+1})$, W_{it+1}/K_{it+1} , δ_{it+1} , and r_{it+1}^B , we winsorize them at the 0% - 95% level. We do not winsorize variables bounded between zero and one, such as $K_{it+1}/(K_{it+1} + W_{it+1})$ or the market leverage, w_{it}^B . Summary statistics and correlation matrix of the aforementioned variables are reported in Table 2 and closely match those in [Gonçalves, Xue, and Zhang \(2020\)](#).

In the model, the fundamental stock return of firm i from year t to $t + 1$, r_{it+1}^F , is constructed with both stock and flow variables at annual frequency. In the estimation, we match r_{it+1}^F with the observed annual return of firm i from the middle of fiscal year t to the middle of fiscal year $t + 1$, following [Gonçalves, Xue, and Zhang \(2020\)](#). Specifically, if firm i ’s fiscal end of year t is month l , r_{it+1}^S , the counterpart of r_{it+1}^F , is the realized 12-month return between month $l - 5$ and $l + 6$. The detailed description about the timing alignment is provided in Section B of the Internet Appendix.

To study anomalies, we construct fundamental portfolio returns based on fundamental firm-level returns. Even though firm-level fundamental returns change annually (in fiscal year), fundamental portfolio returns change monthly because fiscal year-endings vary across firms, and portfolio compositions can also change monthly. However, the fundamental portfolio returns of a

given month are based on annual accounting variables both prior to and after the month. To better align the timing and make a fair comparison, we follow [Gonçalves, Xue, and Zhang \(2020\)](#) and compound the realized portfolio stock returns within a 12-month rolling window with the month in question in the middle of the window. Specifically, we multiply gross returns from month $l - 5$ to month $l + 6$ to match the fundamental returns constructed in month l . Applying this rolling procedure to the realized monthly portfolio returns (January 1967 to June 2017) yields the monthly observations of annualized portfolio returns from June 1967 to December 2016.

We validate our data construction and portfolio formation by successfully reproducing the realized and predicted returns (by the baseline model) of the book-to-market (BM), momentum (R11), asset growth (I/A), and return-on-equity (ROE) deciles in [Gonçalves, Xue, and Zhang \(2020\)](#) using their estimated model parameters. The results are plotted in Figure [A.1](#), which replicates Panel B of Figure 3 in [Gonçalves, Xue, and Zhang \(2020\)](#).

3.2 Estimation methodology

Prior studies ([Liu, Whited, and Zhang, 2009](#); [Gonçalves, Xue, and Zhang, 2020](#), among others) use the General Method of Moments (GMM) to match the unconditional moments derived from equation (6): $E_T[r_{pt+1}^S - r_{pt+1}^F] = 0$ for testing portfolio p , where $E_T[\cdot]$ refers to the operation of taking time series average. To completely avoid portfolio dependence in parameter estimates, we instead target the entire panel of firm-level stock returns using the Bayesian Markov Chain Monte Carlo (MCMC) method. The Bayesian MCMC method can efficiently extract a large amount of information from firm-level stock returns and enable us to allow the parameters to vary across industries and over time. We consider four specifications in the estimation: constant parameters, industry variations only, parameters with time variations only, and parameters with industry and time variations. Next, we explain our methodology in details in terms of the specification that allows both industry and time variations in parameter values.

3.2.1 Bayesian MCMC Denote the technology parameter in the production function for industry j at time t as γ_{jt} and the physical adjustment costs parameter as a_{jt} . The time series of parameter values are referred to as “latent variables” in Bayesian MCMC estimation and are

assumed to evolve as random walk processes:

$$\begin{bmatrix} \gamma_{jt+1} \\ a_{jt+1} \end{bmatrix} = \begin{bmatrix} \gamma_{jt} \\ a_{jt} \end{bmatrix} + \begin{bmatrix} \sigma_\gamma & \\ & \sigma_a \end{bmatrix} \begin{bmatrix} e_{jt+1}^\gamma \\ e_{jt+1}^a \end{bmatrix}, \quad (7)$$

where e_{jt+1}^γ and e_{jt+1}^a follow standard normal distributions independently, and σ_γ and σ_a are the conditional standard deviations of latent variables γ_{jt+1} and a_{jt+1} conditioning on previous time t . Imposing a random walk process on the deep parameters not only encourages persistence, but also enables us to borrow information across time in estimation, leading to more efficient estimates.⁸ For the specification with time variations only, the same random walk process is assumed for all industries, and for the specifications with no time variations, e_{jt+1}^γ and e_{jt+1}^a are set to zero.

Realized stock return of firm i (in industry j) at time $t + 1$ is modeled as the corresponding fundamental return plus an estimation error:

$$r_{it+1}^S = r_{it+1}^F + \varpi_{it}^{-1/2} \sigma_r e_{it+1}^r, \quad (8)$$

where e_{it+1}^r follows the standard normal distribution, σ_r is a parameter to be estimated, and the weight $\varpi_{it}^{-1/2}$ in the estimation error is specified as:

$$\varpi_{it} \equiv \frac{V_{it}}{\sum_{i=1}^{N_{jt}} V_{it}}, \quad (9)$$

in which N_{jt} is the number of firms at time t in industry j to which firm i belongs. By this specification, we introduce heteroskedasticity into the estimation errors of realized stock returns. The variance of a firm's estimation errors decreases with its market equity V_{it} in order to accommodate the fact that stock returns of large firms are less noisy and more reflective of their fundamentals than the returns of small firms.⁹ More importantly, such specification makes the estimated model economically relevant in the sense that it captures the regularity of the majority of the economy. The same rationale motivates the use of NYSE breakpoints in constructing

⁸We also estimate an autoregressive process with order one. The estimated persistence parameters are very close to one for both processes of γ_{jt} and a_{jt} . Thus, we use random walk processes for simplicity.

⁹Large firms have more analysts following than small firms, thus their value is under much closer scrutiny (Bhushan, 1989). Moreover, stocks of large firms are generally more liquid and their market values are less likely to be manipulated or affected by a small group of investors (Amihud, 2002).

portfolios and regressions with weighted least squares in asset pricing studies (e.g., [Hou, Xue, and Zhang, 2015](#)).¹⁰

For the MCMC method, prior distributions of the model parameters need to be specified. We use inverse gamma distributions for the priors of variances: $\sigma_\gamma^2 \sim IG(\kappa_1^\gamma, \kappa_2^\gamma)$, $\sigma_a^2 \sim IG(\kappa_1^a, \kappa_2^a)$, and $\sigma_r^2 \sim IG(\kappa_1^r, \kappa_2^r)$, where κ_1 and κ_2 are hyper-parameters of the inverse gamma distribution (shape-scale parameterizations). The values of κ_1^r , κ_1^γ , and κ_1^a are specified to be 0.01, 1, and 1, respectively; the values of κ_2^r , κ_2^γ , and κ_2^a are chosen to be 0.02, 5, and 5, respectively. The values of κ_1 are chosen relatively small so that the information from data is more likely to dominate (see Section C in the Internet Appendix). The values of κ_2 's are set relatively large so that the variances of the priors are large and thus less informative. Although the choices of these values are seemingly arbitrary, as MCMC runs and information from the data gets entered into the posterior draws, these hyper-parameters weigh less and less. The information from data dominates the posterior draws when MCMC converges.

Finally, the time series of latent variables $\boldsymbol{\theta} \equiv \{\theta_t; t = 1, \dots, T\}$, where $\theta_t = \{(\gamma_{jt}, a_{jt}); j = 1, \dots, 10\}$ and variance parameters $\boldsymbol{\sigma} \equiv \{\sigma_\gamma^2, \sigma_a^2, \sigma_r^2\}$ are drawn in an iterative manner from each complete conditional posterior distribution, resulting in posterior samples from the joint posterior distribution. Based on the model specifications in equations (7) and (8), the joint posterior distribution of $\boldsymbol{\theta}$ and $\boldsymbol{\sigma}$ can be written (in a proportional form) as:

$$\begin{aligned} \mathcal{P}(\boldsymbol{\theta}, \boldsymbol{\sigma} | \mathbf{X}, \mathbf{r}^S, \mathbf{r}^{Ba}) &\propto \prod_{t=0}^{T-1} \prod_{i=1}^{N_{t+1}} \mathcal{N}(r_{it+1}^S; r_{it+1}^F, \sigma_r^2) \times \prod_{t=0}^{T-1} \prod_{j=1}^{N_d} \mathcal{N}(\gamma_{jt+1}; \gamma_{jt}, \sigma_\gamma^2) \\ &\times \prod_{t=0}^{T-1} \prod_{j=1}^{N_d} \mathcal{N}(a_{jt+1}; a_{jt}, \sigma_a^2) \\ &\times \mathcal{IG}(\sigma_r^2; \kappa_1^r, \kappa_2^r) \times \mathcal{IG}(\sigma_\gamma^2; \kappa_1^\gamma, \kappa_2^\gamma) \times \mathcal{IG}(\sigma_a^2; \kappa_1^a, \kappa_2^a), \end{aligned} \quad (10)$$

where N_{t+1} is the number of firms at time $t + 1$, N_d is the number of industries, and r_{it+1}^F is defined in (6). In equation (10), $\mathbf{X} \equiv \{X_{it}; i = 1, \dots, N_t, t = 1, \dots, T\}$ is the panel of fundamental observables, $\mathbf{r}^S$ and $\mathbf{r}^{Ba}$ are the panels of realized stock and bond returns, and $\mathcal{N}(\cdot; \mu, \sigma^2)$ and $\mathcal{IG}(\cdot; \kappa_1, \kappa_2)$ refer to the probability density functions of normal distribution with mean μ and

¹⁰Effort (not reported here) has been made to investigate other kinds of functional forms relating the variability in estimation errors of a firm's stock returns to its market equity. The relationship specified in equation (9) best fits the data in terms of mean absolute error (*m.a.e.*) of firm-level stock returns.

variance σ^2 and inverse gamma distribution with shape-scale parameters κ_1 and κ_2 , respectively. We run 20,000 MCMC iterations and use the last 5,000 iterations to obtain posterior draws. We confirm the convergence of the posterior distributions. Section C in the Internet Appendix details the sampling algorithm and posterior derivations.

3.2.2 Comparison of Bayesian MCMC with GMM and NLS Our MCMC estimation approach is fundamentally different from the estimation method in Liu, Whited, and Zhang (2009) and Gonçalves, Xue, and Zhang (2020), among others, and it offers several advantages. First, our estimates of parameter values are independent of any specific testing portfolios. We utilize the entire *distribution* of firm-level stock returns to estimate model parameters, while the GMM in Liu, Whited, and Zhang (2009) and Gonçalves, Xue, and Zhang (2020) matches the time-series averages of returns on testing portfolios. This feature is critical for addressing the critique of Campbell (2017) that the parameter values of the model are chosen to fit a specific set of anomalies and different values are required for different anomalies.

Second, our MCMC algorithm generates random draws of model parameters from their joint posterior distribution given the observations on firms' stock and bond returns and fundamentals, while GMM outputs point estimates of model parameters, which are *deterministic* given the same set of observations. One advantage of our Bayesian approach is that *probabilistic* inferences for the estimated parameters and fundamental stock returns can be easily made using posterior draws from the MCMC iterations.

Third, given the vast amount of information in firm-level stock returns exploited by our Bayesian MCMC estimation, we are able to accurately identify the true values of model parameters even for the specification where these parameters vary across industries and over time. This feature can be extremely important when the modeled economy is highly heterogeneous and changing over time. With the Fama-French 10-industry classification, we estimate 2×500 latent variables for the sample between 1967 to 2016. Although the number of latent variables is small compared to the number of observations used in our estimation (136,598), it is extremely difficult to estimate them via either GMM, or a maximum likelihood based approach. Prior studies thus assume time-invariant parameter values in general.

Lastly, the frequentist method of Nonlinear Least Squares (NLS) can also be used to estimate

industry-specific and time-varying parameters using firm-level stock returns as follows:

$$\hat{\theta}_{jt+1}^{NLS} = \arg \min_{\theta_{jt+1}} \sum_{i=1}^{N_{jt+1}} \varpi_{it} \left[f \left(X_{it}, X_{it+1} | \hat{\theta}_{jt}^{NLS}, \theta_{jt+1} \right) - r_{it+1}^S \right]^2, \quad (11)$$

where N_{jt+1} is the number of firms in industry j at time $t+1$, $\hat{\theta}_{jt}^{NLS}$ is the estimated parameters for industry j at t , and ϖ_{it-1} , which is proportional to the market equity V_{it-1} as defined in equation (9). However, only the information of industry j at time t is used to identify $\hat{\theta}_{jt+1}^{NLS}$ s. In contrast, the posterior of an industry-time specific parameter, θ_{jt} , in Bayesian MCMC utilizes the information of the entire data sample. The reasons are as follows. First, the random walk process imposed on parameters in equation (7) connects information across different points in time. Second, physical adjustment costs parameter a_{jt} enters into the probability distributions of both r_{it} and r_{it+1} of firm i in industry j as shown in equations (2) and (10), which also connects information in returns across time. Third, due to entry and exit, the probability distribution of stock return r_{it+1} can also connect information in returns across industries if firm i switches from industry j to k at time $t+1$. Consequently, the identification of any specific latent variables a_{jt} and γ_{jt} utilizes the information of the entire data sample.¹¹

We use simulation studies to examine whether Bayesian MCMC can discover the true parameter values under our model framework, which is highly nonlinear, and also compare the performance of Bayesian MCMC and NLS.¹² The results show that Bayesian MCMC performs very well for our highly nonlinear model and is able to discover the true parameter values under all four specifications, while NLS often fails to discover the true parameter values that are time-varying. The details of the simulation studies are included in Section D of the Internet Appendix.

Finally, we conduct an extensive set of simulation studies to investigate the performance of Bayesian MCMC when the model is misspecified and compare its estimation biases with those of the GMM estimation targeting the average returns of testing portfolios. Section F of the Internet Appendix discusses the situation when firms make investment decisions monthly instead of annually and Section G discusses the situation when the estimated model missed an additional capital in

¹¹Even though γ_{jt} enters into the probability distribution of stock return r_{it} only for firm i in industry j at time t , γ_{jt} is identified together with a_{jt} . Therefore, the value of γ_{jt} also reflects the information of the entire data sample.

¹²Detailed discussion on the differences between Bayesian and NLS, and the comparison of their estimation accuracy under our model framework are provided in Section E of the Internet Appendix.

the production function. Not surprisingly, neither Bayesian MCMC nor GMM can discover the true parameter values when the model is misspecified. More importantly, in neither case do we find GMM outperform Bayesian MCMC in terms of the magnitude of the estimation biases. We suspect that given the highly nonlinear relation between return and firm fundamentals, averaging returns in the cross section and time series as in GMM is unlikely to eliminate these model misspecification errors.

4 Parameter Estimation and Performance Evaluation

We conduct estimations of four model specifications: constant parameters, industry-specific parameters, time-varying parameters, and industry-specific and time-varying parameters. We compare our estimates with those in [Gonçalves, Xue, and Zhang \(2020\)](#) and discuss whether the estimated parameter values make economic sense. Finally, we compare the performance of these four specifications in terms of matching firm-level stock returns.

4.1 Parameter estimates

Our estimation generates the posterior distributions of the marginal product parameter γ and the physical investment adjustment costs parameter a . For each parameter estimate, we report its posterior mean and credible interval (CI), the latter of which refers to the interval wherein a posterior draw of the parameter falls with the specified probability. A small CI indicates that the parameter is accurately estimated. ¹³

Panel A of Table 3 shows that under the specification with constant parameters, the posterior mean of γ is 0.15 with a CI of [0.15,0.15] and the posterior mean of a is 0.14 with a CI of [0.13,0.14]. These narrow CIs indicate that the parameter values are identified with high precision due to the simple model structure (only two structural parameters) and the large amount of information (136,598 firm-level observations). Panels B to D present estimation results under the specifications with industry-specific parameters, time-varying (industry invariant)

¹³Formally, the posterior credible interval I_p of parameter θ satisfies $P(\theta \in I_p | \mathbf{X}, \mathbf{r}^S, \mathbf{r}^{Ba}) = p$, where p is the probability. A credible interval is the Bayesian equivalent of the confidence interval in frequentist statistics. A credible interval is the Bayesian counterpart of the confidence interval in frequentist statistics. Confidence intervals treat the estimated parameter as a fixed value and the bounds are random variables due to random sampling, whereas credible intervals treat the parameter as a random variable and the credible bounds are determined by the derived posterior distribution.

parameters, and industry-specific and time-varying parameters, respectively. For specifications with time-varying parameters (Panels C and D), we report the time-series averages of the posterior means and 95% credible intervals of γ and a and their time-series standard deviations under Columns $\sigma(\gamma)$ and $\sigma(a)$.

Several observations emerge from Table 3. First, parameters are accurately estimated with narrow CI's even when parameters vary over time and across industries. This result highlights that the Bayesian MCMC is able to extract a large amount of information from firm-level stock returns and precisely identify the parameter values even when they are industry-specific and time-varying.

Second, allowing time variation in parameters is crucial for getting unbiased estimates when the underlying model is nonlinear in these parameters. When time variation is shut down, the estimates of γ (in Panels A and B) are close to the time series average of their time-varying counterparts (in Panels C and D). In contrast, we see large changes in the estimates of a when time variation is shut down. For example, the posterior mean of γ for Business Equipment sector is 0.23 in Panel B, which is the same as its time-series average in Panel D. However, the posterior mean of a for Business Equipment sector is 0.78 in Panel B, less than half of its time-series average of 1.78 in Panel D. The reason is that model-implied return is a linear function of γ but a nonlinear function of a as shown in equation (6). If the true value of the parameter is time-varying, when information is aggregated along the time dimension, a nonlinear relation used for the estimation often generates an estimate that is far away from its time-series average.¹⁴

Third, the parameter estimates differ greatly across industries. For example, under the specification with industry and time variations (Panel D), γ is estimated to be 0.08 on average with a CI of [0.07, 0.09] for the wholesale & retail sector, compared to 0.28 for the telecom sector,

¹⁴There are three layers of aggregation in existing estimation methods. The first layer of aggregation is to aggregate firm-level fundamentals, such as investment and profits, to get portfolio-level fundamentals, which are used as the inputs to compute portfolio-level model-implied fundamental returns. Given that the model prediction is only valid at the firm level, this aggregation can introduce significant biases into parameter estimates. [Gonçalves, Xue, and Zhang \(2020\)](#) propose a method to avoid this aggregation. They directly use firm-level fundamentals to compute firm-level model-implied returns.

The second and third layers of aggregation occur when the estimation tries to match portfolio-level return moments. To compute portfolio returns, firm-level returns are aggregated based on certain firm characteristics such as size and book-to-market ratio, which is the second layer of aggregation and is at the center of the [Campbell \(2017\)](#)'s critique. Lastly, the third layer of aggregation happens when the estimation targets at the time-series average (or other statistics) of portfolio returns, in which returns are aggregated across time. [Belo et al. \(2022\)](#) point out that the third layer of aggregation can lose relevant information for the identification of the model parameters. They thus target the entire time series of the mean valuation ratio (their variable of interest) of a portfolio to avoid the first and third layers of aggregation. We go one step further to target the entire panel of firm-level returns.

consistent with the fact that capital is less important for the wholesale & retail sector than for the telecom sector. The average posterior mean of a ranges from 0.25 for Utilities to 1.78 for Business Equipment. Table 3 shows that the business equipment sector has the highest marginal q of physical capital, q^K , of 1.53 while the utilities sector has the lowest q^K of 1.02. Hence, these estimates of a are consistent with our intuition that the business equipment sector, which includes the high-tech firms, has the highest growth potential while the regulated utility sector has the lowest potential for growth. Similar cross-industry variations in parameter values can be observed from Panel B, which reports the estimation results under the specification with industry-specific but time invariant parameters.

Overall, Table 3 shows that there are large variations in parameter values across industries and over time.¹⁵ Estimations that fail to recognize heterogeneities in parameter values could lead to vastly different estimates from the sample average of their true values if the underlying model is nonlinear. Bayesian MCMC is a useful tool for such scenarios. The value of a turns out to be critical for the model to generate the value premium, and we will provide a detailed discussion later in Section 6. We also conduct empirical tests on the model-predicted relation between firm-level profit margin and the estimated industry-specific and time-varying γ and similarly the relation between firm-level Tobin's q and the estimated a . It is important to note that we do not expect these relations in the data to be exactly the same as what the model predicts because the model is extremely simplified to derive the Hayashi (1982) equality between stock return and firm fundamentals. The results show that γ and a have statistically significant relation with firms' profit margin and q , which indicates there are relevant economic contents in our parameter estimates. The detailed discussions are included in Section H of the Internet Appendix.

4.2 Overall fit of the estimation

In this subsection, we compare a set of key moments of the realized and fundamental firm-level stock returns to gauge the overall fit of the four estimation specifications. Table 4 reports the posterior means and 95% credible intervals of the mean, standard deviation, skewness, kurtosis,

¹⁵Parameter variation over time could also originate from time misalignment. Due to data limitation, our estimation assumes that firms make investment decisions at annual frequency. Section F of the Internet Appendix shows that when firms make investment decisions monthly instead of annually, even if the monthly parameters are constant, the true annual parameters are time-varying. Bayesian MCMC thus provides a reasonable reduced-form way to deal with this potential model mispecifications.

the time-series average of the cross-sectional correlations between fundamental and realized stock returns, and the mean absolute error (m.a.e.) for the specifications with constant, industry-specific, time-varying, and industry-specific and time-varying parameter values.¹⁶ The m.a.e. is defined as

$$\text{m.a.e.} \equiv \frac{1}{T} \sum_{t=0}^{T-1} \frac{1}{N_{t+1}} \sum_{i=1}^{N_{t+1}} |r_{it}^S - r_{it}^F|, \quad (12)$$

where N_{t+1} is the number of firms in period $t + 1$, and r^S and r^F are the realized and fundamental stock returns, respectively. The same moments of the realized stock returns are presented for comparison.

Several observations emerge from Table 4. First, the mean, skewness and kurtosis of the fundamental returns match well with those of the realized returns across all four specifications. The mean of fundamental stock returns ranges from 14.97% to 15.65% across specifications, compared to 14.45% in the data. The skewness ranges from 1.66 to 2.12, compared to 2.15 in the data, and kurtosis ranges from 10.66 to 13.33 compared to 11.05 in the data. Second, the standard deviations of the fundamental returns are much smaller than that of the realized ones. The specification with industry-specific and time-varying parameters generates the highest standard deviation 34.17%, compared to 60.78% in the data, while the specification with industry-specific parameters generates the lowest standard deviation, 18.49%. Third, the time-series average of the cross-sectional correlations between realized and fundamental firm-level stock returns is highest for the specification with industry-specific and time-varying parameter values (0.20), and lowest for the specifications with constant parameter values (0.09). Lastly, the m.a.e. is lowest for the specification with industry-specific and time-varying parameter values (40.10%), followed in turn by the specifications with time-varying (40.85%), industry-specific (41.85%), and constant parameter values (42.45%).¹⁷

Figure 1 plots the histograms of realized (in blue) and fundamental (in orange) firm-level returns based on the posterior means of the parameter estimates under the four specifications. Consistent with what Table 4 shows, realized returns have a much wider distribution than fundamental returns

¹⁶Given firm-level accounting variables, each posterior draw of $\theta = \{\theta_t; t = 1, \dots, T\}$ leads to a panel of firm-level fundamental returns and any statistical moments of these returns.

¹⁷For comparison, we report the same statistics of the NLS estimation for the four specifications in Table A.3 in the Internet Appendix. Bayesian MCMC results in a smaller m.a.e. in every specification, echoing the superior performance of Bayesian approach compared to NLS documented in Sections D and E of the Internet Appendix.

at both left and right tails and thus have a larger standard deviation. Both the realized and the four fundamental distributions have longer right tails, resulting in positive skewness and kurtosis larger than 3. Next, we conduct a rigorous comparison in performance among the four specifications.

4.3 Performance comparison of the four specifications

We evaluate the performance of each specification based on the mean absolute error (m.a.e.) of the fundamental firm-level stock returns. Different from GMM, which gives point estimates of the parameters, Bayesian MCMC offers a probabilistic view of the parameters and thus the fundamental returns. Given firm-level accounting variables, each posterior draw of $\boldsymbol{\theta} = \{\theta_t; t = 1, \dots, T\}$ leads to a panel of firm-level fundamental returns. Therefore, our estimation generates posterior distributions of the fundamental returns and any statistical moments of these returns. Based on the posterior distribution of the m.a.e. under each specification, we test whether the differences in m.a.e. of the four specifications are statistically significant using the following statistic:

$$d^a = \frac{1}{T} \sum_{t=0}^{T-1} \frac{1}{N_{t+1}} \sum_{i=1}^{N_{t+1}} \left(\left| r_{it+1}^S - r_{it+1}^{F(a)} \right| - \left| r_{it+1}^S - r_{it+1}^{F(b)} \right| \right), \quad (13)$$

where $r^{F(b)}$ and $r^{F(a)}$ are the fundamental returns under the benchmark specification, defined as the one with the lowest m.a.e. (i.e., specification with industry-specific and time-varying parameters), and under an alternative specification, a , respectively. Note that although our statistic is similar in form to [Diebold and Mariano \(1995\)](#), they differ in nature by statistical properties. However, taking advantage of MCMC, we can still make a valid inference from this statistic. Intuitively, when the estimation errors from the alternative specification are larger in magnitude, we expect d^a to be above 0 with statistical significance. Following equation (13), we record $d^{a(m)}$ for the m -th posterior draws of parameter values. These 5,000 posterior draws jointly provide us with the empirical distribution of the statistic d^a . A significantly positive d^a indicates that the benchmark specification performs significantly better than the alternative specification a in explaining firm-level stock returns.

Figure 2 plots the distribution of d^a for each of the three alternative specifications all in one panel in Panel (a) and separately in Panels (b) to (d). The 2.5, 50, and 97.5 percentiles of the posterior distributions are also marked on the x-axis in the last three panels. We can see that

the posterior distributions of these three settings all lie in the positive region. Neither of the three 95% credible intervals ($[1.80, 1.90]$, $[1.22, 1.32]$, and $[0.57, 0.67]$ for the specifications with constant, industry-specific, and time-varying parameter values) includes zero, indicating that the specification with industry-specific and time-varying parameters performs significantly better in explaining the firm-level stock returns than these alternative specifications. Moreover, the location of these distributions in Panel (a) indicates that the performance of the model with time-varying parameters is closest to the benchmark, followed by the settings with industry-specific parameters and with constant parameters.

In sum, the specification with industry-specific and time-varying parameters has significantly better performance in matching firm-level stock returns than the other three specifications. In addition, time variation is more crucial to the superior performance than industry variation. Next, we explore the capability of the model in generating anomaly return spreads and matching the dynamics of anomaly portfolios.

5 Fundamental Anomalies

In this section, we examine whether the model-implied fundamental returns exhibit the same return regularities, in terms of the average anomaly premiums and their time-series dynamics, when the model parameters are estimated to match firm-level stock returns. We focus on a set of well-documented and robust anomalies based on the replication study by [Hou, Xue, and Zhang \(2020\)](#), who replicate 452 anomalies, classified into six categories: momentum, value versus growth, investment, profitability, intangibles, and trading frictions. They show that for the sample period as ours, from June 1967 to December 2017, only 153 anomalies can be successfully replicated.¹⁸ We then choose 12 anomalies, covering all six categories, that are most widely studied in the literature and can be successfully replicated in our sample period. More importantly, investment-based asset pricing models have been suggested by prior studies to explain these anomalies: value anomaly sorted on book-to-market equity ratio (BM); momentum anomaly sorted on the prior 11-month returns skipping the most recent month (R11); four investment anomalies sorted on asset growth (I/A), net stock issues (NSI), investment-to-assets

¹⁸The replication of an anomaly is successful if the average return of its high-minus-low decile is significant at the 5% threshold based on portfolio sorts with NYSE breakpoints and value-weighted returns.

ratio ($\Delta\text{PI}/A$), and accruals (Accruals); three profitability anomalies sorted on return-on-equity (ROE), return-on-assets (ROA), and gross profitability (GP/A); two intangibles anomalies sorted on R&D expenses-to-market ratio (RD/M) and advertising expenses-to-market ratio (Ad/M); and one trading frictions anomaly sorted on market capitalization (Size).¹⁹

5.1 Anomaly premiums

Figure 3 plots the posterior distributions of the 12 fundamental factor premiums under the specification with industry-specific and time-varying parameters and labels the 2.5, 50, and 97.5 percentiles of each distribution. For example, the posterior distribution in Panel “BM” indicates that, given the observed accounting variables and provided that the model is correctly specified, the fundamental value premium per annum falls in the range between 0.31% and 0.60% with 95% probability and the posterior median is 0.46% per annum. The red line in each panel presents the density function of a normal distribution with mean and standard deviation taken from the corresponding posterior distribution. Notice that the posterior distributions are very much close to normal distribution, indicating that our Bayesian MCMC algorithm converges well. In cases where the Bayesian MCMC algorithm does not converge, the posterior distributions typically would have multiple peaks or/and long and fat tails. Moreover, the credible intervals of the fundamental factor premiums are extremely narrow. As we argue before, these tight posterior distributions indicate that the simplicity of the model and the richness of firm-level information confine the parameter estimates and the fundamental factor premiums to a small set of possible values.

Table 5 presents the fundamental (value-weighted) factor premiums and alphas, defined as the difference between realized and fundamental factor premiums, i.e., $\alpha \equiv r^S - r^F$, of the high minus low hedging portfolios for each of the 12 anomalies under the four specifications. Their t -values are reported in parentheses. The results show that the specification with industry-specific and time-varying parameters (θ_{jt}) has the best performance in terms of generating factor premiums that are consistent with what is observed in the data, followed in turn by the one with industry-specific parameters (θ_j), the one with time-varying parameters (θ_t), and the one with constant parameters

¹⁹Hou, Xue, and Zhang (2020) show that almost all well-known anomalies in the trading frictions category cannot be successfully replicated. For completeness, we include the Size anomaly from this category since it is one of the most widely studied anomalies in the literature.

(θ). Specifically, all four specifications are able to generate significant, at least at the 10% level, momentum (R11), profitability (ROE, GP/A, and ROA), size, investment (NSI and $\Delta\text{PI}/A$), and advertising (Ad/M) premiums with the correct signs. However, without industry variations, the model cannot generate significant RD/M premium. And both industry and time variations in parameters are needed to generate significant I/A premium. Finally, none of the specifications can generate a significant value premium and the accruals anomaly with the correct sign.

A significant alpha indicates that the fundamental factor premium is significantly different from the realized one in magnitude. Specification θ_{jt} fails to match the magnitudes of the I/A, NSI, GP/A, value, and accrual premiums. Additionally, without time variations, specification θ_j cannot match the momentum premium, and without industry variations, specification θ_t fails to match the momentum and intangibles (RD/M and Ad/M) premiums. Finally, the specification with constant parameter values θ can only match the magnitudes of the ROE, Size, NSI, and ROA premiums.

Overall, the estimated models under all specifications successfully generate significant momentum, size, profitability, and investment premiums. However, even the best-performing specification, θ_{jt} , fails to generate a significant value premium. This contrasts sharply with the findings of [Gonçalves, Xue, and Zhang \(2020\)](#), whose estimation is based on the same model and same data sample as ours. The difference is that the ten value deciles are among the 40 testing portfolios in [Gonçalves, Xue, and Zhang’s \(2020\)](#) estimation. We argue that this inclusion overstates the model’s ability to explain these anomalies. Our results indicate that [Campbell’s \(2017\)](#) critique remains a concern and underscores the importance of using portfolio-independent parameters to evaluate a model’s capability in generating stock market anomalies.

5.2 Correlation between realized and fundamental portfolio returns

Table 6 reports the contemporaneous correlations between the realized and fundamental returns on the 120 decile portfolios and the 12 high-minus-low portfolios for the 12 anomalies under the four specifications. Not surprisingly, specification θ_{jt} performs the best, under which the fundamental and realized portfolio returns are all highly correlated and the correlation coefficients are all significant at the 1% level. The average correlation is 0.69 for decile portfolios and 0.43 for the high-minus-low portfolios.

The comparison among the four specifications indicates that both the industry and time

variations in parameter values are critical to the high correlations for the high-minus-low portfolio returns, while only time variation is critical to the high correlations for the decile portfolios. Specifically, the average correlation of the high-minus-low portfolio returns, $\bar{\rho}_{H-L}$, is 0.43 under θ_{jt} . It drops to 0.18 under θ_j when time variation is shut down, to 0.13 under θ_t when industry variation is shut down, and to 0.12 when both time and industry variations are shut down. At the same time, the fraction of positive and significant decile correlations drops from 100% under θ_{jt} to 67% under θ_j , 58% under θ_t , and 50% under θ .

On the other hand, the average correlations of the decile portfolio returns, $\bar{\rho}$, drops from 0.69 under θ_{jt} to 0.66 under θ_t when industry variation is shut down, to 0.06 under θ_j when time variation is shut down, and to 0.03 under θ when both time and industry variation are shut down. Similarly, the fraction of positive and significant decile correlations drops from 100% under θ_{jt} and θ_t to 47.5% under θ_j , and 42.5% under θ .

Note that [Liu, Whited, and Zhang \(2009\)](#) document a correlation puzzle that the contemporaneous correlations between the realized and fundamental returns are weakly negative. [Gonçalves, Xue, and Zhang \(2020\)](#) show that the correlation turns positive under their estimated model.²⁰ Our estimated models generate higher correlations than the ones in [Gonçalves, Xue, and Zhang \(2020\)](#). Among the 40 BM, R11, I/A, and ROE decile portfolios studied by [Gonçalves, Xue, and Zhang \(2020\)](#), 7 out of 40 correlations are positive and significant (at least at the 10% level) in Panel B of their Table 6, compared to 19 out of 40 under our specification θ . With industry and time variations in parameter values, our estimations yield even higher correlations, as demonstrated above. Therefore, the high correlations in Table 6 stem from the target of our estimation method—the entire sample of firm-level returns—and its flexibility to allow industry and time variations in parameter values.

²⁰[Gonçalves, Xue, and Zhang \(2020\)](#) argue that the reason behind the positive correlation is the better alignment between the timing of realized and fundamental returns. They use a rolling procedure to construct the annualized monthly portfolio returns, which we follow in our paper and is explained in Section 3.1.2. We show in the unreported work that this rolling procedure is not the reason why our model generates such high correlations. Following [Liu, Whited, and Zhang \(2009\)](#), we restrict our sample to the one with December fiscal year end and thus do not need to use the rolling procedure for timing alignment. The resulting correlations remain as high as the reported ones in Table 6.

6 Discussions

The small estimated value of a is the reason why our model fails to generate the value premium, whereas the model in [Gonçalves, Xue, and Zhang \(2020\)](#) successfully does. Although both use the same model and data sample and construct fundamental returns at the firm level, we estimate parameter values by matching each firm’s annual stock returns for the entire sample from 1967 to 2016, while [Gonçalves, Xue, and Zhang \(2020\)](#) target the average returns of specific testing portfolios. Our estimated investment adjustment costs parameter a is substantially smaller than the ones in [Gonçalves, Xue, and Zhang \(2020\)](#).²¹ For example, [Gonçalves, Xue, and Zhang \(2020\)](#) estimate that a is 2.84 (std = 0.47) in their baseline using GMM that targets the average returns of the value, momentum, investment (I/A), and profitability (ROE) decile portfolios. In contrast, our estimate under the specification of constant parameter values is 0.14.

Table [A.1](#) in the Internet Appendix shows that value firms have lower lagged investment-to-capital and sales-to-capital ratios than growth firms do. At the same time, equation (6) implies a negative relation of stock return to I_{it}/K_{it} but a positive relation to Y_{it+1}/K_{it+1} . Consequently, for the model to generate a positive and significant value premium, the investment adjustment costs parameter needs to be large enough. The question is how reasonable are our estimates of a ? And more importantly, are the low estimates of a at odds with the values used in the investment-based asset pricing models to generate the value premium? We aim to answer these questions in this section. At the end, we also discuss possible reasons why the model fails to account for the accrual anomaly.

6.1 The value anomaly

6.1.1 Values of a in the literature to generate the value premium Our model features a symmetric adjustment cost function to ensure [Hayashi \(1982\)](#) equality. Moreover, given that 92.02% (92.06%) of the observations in our sample are positive (non-negative) investments and the estimation is based on annual data, our estimate is essentially the parameter value for positive investment at annual frequency. Most theoretical models in the literature are calibrated (or

²¹Their parameter estimates vary with the testing portfolios under their baseline estimation method, ranging from 1.63 to 8.11 as shown in Panel B of Table 3 of [Gonçalves, Xue, and Zhang \(2020\)](#).

estimated using SMM) at monthly frequency with the exception of [Tuzel \(2010\)](#) and [Li \(2017\)](#). Panel A of Table [A.11](#) in the Internet Appendix lists 13 theoretical articles in asset pricing published since 1999 that feature quadratic investment adjustment costs. For each article, the table also presents the calibrated adjustment cost parameter values, $a+$ for positive investment and $a-$ for negative investment. All parameter values are converted into annual values for comparison.²² Apparently, a wide range of values have been used for $a+$ in the literature, with 0.00 and 0.08 in the low end and 25 in the high end.

For comparison, the baseline estimate of a in [Gonçalves, Xue, and Zhang \(2020\)](#) is 2.84, which is on the higher end of the range: Only two papers ([Bai et al., 2019](#); [Herskovic, Kind, and Kung, 2023](#)), out of the 13 papers in Table [A.11](#), use a value of a larger than [Gonçalves, Xue, and Zhang \(2020\)](#)'s estimate. On the other hand, our estimate of a is 0.14 under the specification of constant parameters, 0.19 on average under the specification of industry-specific parameters, 0.29 on average under the specification of time-varying parameters, and 0.727 on average under the specification of industry-specific and time-varying parameters. These values lie on the low end of, but still within, the range of the values used in the literature. Even for our lowest estimate (0.14), there are two papers ([Li, 2017](#); [Belo, Lin, and Yang, 2019](#)) use values lower than that. Therefore, although our estimates are significantly lower than the estimates in GXZ, they are within the range of values used in the investment-based asset pricing models.

In addition, in a widely cited study with the sole purpose of estimating the adjustment cost parameter, [Cooper and Haltiwanger \(2006\)](#) estimate that the quadratic adjustment cost parameter is 0.455 when only quadratic adjustment cost is present and the value drops to 0.049 when non-convex adjustment costs are added, by matching the serial correlation of investment rates, correlation of profit shocks and investment rates, and positive and negative investment spike rates.²³ Both of these estimates are closer to our estimates than to those in [Gonçalves, Xue, and Zhang \(2020\)](#). More importantly, using simulated data, [Cooper and Haltiwanger \(2006\)](#) show that due to measurement errors in average q , the coefficient on average q in investment- q regressions implies an estimate of a 100 times as large as its true value. This study explains why

²²The sum of 12 monthly adjustment costs within a given year is by definition the annual investment adjustment cost. Assuming the annual investment rate is on average 12 times the monthly one, the annual adjustment costs parameter is one twelfth of the monthly costs parameter under the quadratic cost function.

²³[Cooper and Haltiwanger \(2006\)](#) define episodes of investment rates in excess of 20% spikes.

prior estimates based on investment- q regressions generate extremely large estimates a (for example, 20 in Hayashi (1982) and 3 in Gilchrist and Himmelberg (1995)). After correcting the measurement error problem using instrumental variables, Hall (2004) finds the estimates of a close to zero (especially the ones with high precision as shown in his Table III) and statistically insignificant. Thus, although our estimates of a are lower than those in Gonçalves, Xue, and Zhang (2020), they are comparable to other ones in the literature.

6.1.2 Mechanism to generate the value premium Asset pricing models that generate the value premium often rely on costly investment reversibility. Table A.11 lists 28 such theoretical articles in asset pricing that have been published since 1999. All feature costly reversibility in their models, including investment irreversibility, asymmetric nonconvex and/or convex adjustment cost, or lower resale price of capital. Panel A lists papers that feature quadratic adjustment costs and the calibrated or estimated values of these cost parameters. Costly reversibility makes value firms riskier than growth firms during bad times, resulting in a higher risk premium for value firms. Therefore, the value premium is primarily driven by the large adjustment cost of negative investment (Zhang (2005), among others).

Bai et al. (2023) estimate, via SMM, a model with asymmetric investment adjustment cost to match the time series averages of the cross-sectional standard deviation, skewness, autocorrelation, and the fraction of negative investment rates of the annual gross investment rates, the time series averages of cross-sectional standard deviation and skewness of annual stock excess returns, and the average value premium in monthly percent. In Table S3 of Bai et al. (2023), the estimated monthly cost parameter for positive investment is 0.1385, equivalent to an annual cost parameter of 0.012) and statistically insignificant. On the contrary, the cost parameter for negative investment is 102.26 (annual value of 8.52) and highly significant. Under this set of parameters, the model is able to generate a large and positive value premium. However, once the cost parameter for negative investments is lowered to the same value as the cost parameter for positive investments, the value premium becomes negative.

Similarly, Favilukis and Lin (2016) calibrate a symmetric quarterly adjustment cost function with a parameter value ranging from 0.3 (equivalent to a monthly value from 0.075) under the Cobb-Douglas production function to match the volatility of investment rate. They show in Table

6 of the paper that the model cannot generate the value premium with symmetric adjustment costs, unless they add wage rigidity to the model, which acts as the operational leverage. In addition, they show that higher adjustment costs always help to increase equity volatility and the value premium, but they decrease aggregate investment volatility.

In sum, our model is highly simplified, and we do not expect it to generate all the anomalies observed in the data. The central argument of our paper is that a model’s ability to explain anomalies can be overstated if its parameters are selected to match the anomaly returns it aims to explain. The failure to generate the value premium when the average returns of the book-to-market deciles are not the target moments illustrates this precisely. Our results indicate that [Campbell \(2017\)](#)’s critique continues to be a serious concern even if portfolio returns are aggregated from model-implied firm-level stock returns as in [Gonçalves, Xue, and Zhang \(2020\)](#).

6.2 The accruals anomaly

Table [A.14](#) of the Internet Appendix shows that Y_{it+1}/K_{it+1} is the most important driver of the return spreads across firms sorted on accruals. According to Table [A.1](#) in the Internet Appendix, high-accruals firms have higher Y_{it+1}/K_{it+1} than low-accruals firms do, which leads to a positive high-minus-low accruals spread, opposite of the one in the data.

However, differences in Y_{it+1}/K_{it+1} between high- and low-accruals firms are likely overstated. The concept of accruals is absent in our model and cash and accruals basis accountings are treated the same. The earnings of the high- and low-accruals firms are assumed to have the same quality. High accruals mean high profitability in the model, which however is not necessarily true in the data. Data shows that subsequent write-offs of account receivables happen more often to high-accruals firms ([Dechow and Dichev, 2002](#), among others). Therefore, the true difference in profitability between high- and low-accruals firms is likely much smaller, which may lead to a smaller or even negative fundamental accruals premium.

Prior literature, such as [Wu, Zhang, and Zhang \(2010\)](#) and [Zhang \(2007\)](#) among others, argues that high-accruals firms have lower average returns due to their higher investment rate on working capital. The intuition is analogous to that of the physical investment premium. Assuming that investment in working capital incurs adjustment costs, a higher investment rate on working capital leads to a higher marginal cost of working capital investment and thus lower fundamental stock

return. Our baseline model assumes zero adjustment costs on working capital investment for comparison with [Gonçalves, Xue, and Zhang \(2020\)](#). We thus estimate an extended model with quadratic adjustment costs on working capital investment and details are explained in Section [A.3](#) of the Internet Appendix. Parameter estimates and the corresponding fundamental anomaly premiums are reported in Table [A.12](#) and Table [A.13](#), respectively. The results are quantitatively similar to the baseline ones and specifically, the accruals premium stays positive and significant. These results indicate that the failure to generate accruals premium is not due to the absence of adjustment costs on working capital investment.²⁴ Overall, our evidence suggests that explicitly modeling earnings quality is a more promising direction to generate the accruals anomaly, which we leave for future research.

7 Conclusion

Can stock market anomalies be replicated within an investment-based asset pricing framework? To answer this question, prior studies often choose model parameters to fit the same set of anomaly returns that these studies aim to explain. In this paper, we propose a portfolio-independent estimation methodology based on Bayesian MCMC, which can be used to fairly evaluate the performance of any given model in explaining anomalies.

Applying this method to the same two-capital q -model as that in [Gonçalves, Xue, and Zhang \(2020\)](#) to match the entire sample of firm-level returns, our estimations yield higher correlations between realized and fundamental portfolio returns compared to prior literature. The estimated two-capital model also generates large and significant size, momentum, profitability, investment, and intangibles premiums.

However, we show that even under our best fitting estimation with industry-specific and time-varying parameter values, the model fails to generate the value premium. Our results thus echo [Campbell \(2017\)](#)'s critique and question the practice of evaluating a model's capability in generating certain anomaly premiums based on the set of model parameters chosen to match the corresponding decile portfolio returns.

²⁴Our estimates of the adjustment cost parameter on working capital investment are close to the estimates in [Gonçalves, Xue, and Zhang \(2020\)](#). When using [Gonçalves, Xue, and Zhang \(2020\)](#)'s baseline parameters estimates for the extended model with adjustment costs on working capital investment (reported in Section E of Internet Appendix), we also find positive implied accruals premiums as well.

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Appendix

A Definition of Sorting Variables

BM (Davis, Fama, and French, 2000) Book-to-market equity ratio, defined as the book value of equity for fiscal year end in the previous calendar year $t - 1$ divided by the market value of equity at the end of December of the previous calendar year $t - 1$. We measure book equity as stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (item TXDITC or the sum of item TXDB and item ITCB) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ) if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

R11 (Fama and French, 1996; Carhart, 1997) Prior 11-month returns from month $t-12$ to $t-2$.

I/A (Cooper, Gulen, and Schill, 2008) We measure I/A as change in total assets (Compustat annual item AT) scaled by lagged total assets. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on I/A for the fiscal year ending in calendar year $t-1$ and calculate monthly value-weighted decile returns from July of year t to June of $t+1$.

ROE (Hou, Xue, and Zhang, 2020) ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged book equity. From 1972 onward, quarterly book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock (item PSTKQ). Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the book value of preferred stock (item PSTKQ), or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity. Prior to 1972, we expand the sample coverage by using book equity from Compustat annual files and imputing quarterly book equity with clean surplus accounting.

At the beginning of each month t , we sort stocks into deciles on their most recent ROE. Before 1972, we use the most recent ROE computed with quarterly earnings from the fiscal quarter ending at least four months ago. From 1972 onward, we use ROE computed with quarterly earnings from the most recent quarterly earnings announcement date (item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter corresponding to its most recent ROE to be within six months prior to the portfolio formation and its earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t+1$.

Size (Fama and French, 1992) Size is price times shares outstanding from CRSP. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on the June-end *Size*, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$.

Accruals (Sloan, 1996) We measure Accruals as $\frac{\Delta ACT - \Delta CHE - \Delta LCT + \Delta DLC + \Delta TXP - \Delta DP}{(AT + AT_{-1})/2}$, where ΔACT is the annual change in total current assets, ΔCHE is the annual change in total cash and short-term investments, ΔLCT is the annual change in current liabilities, ΔDLC is the annual

change in debt in current liabilities, ΔTXP is the annual change in income taxes payable, ΔDP is the annual change in depreciation and amortization, and $(AT + AT_{-1})/2$ is average total assets over the last two years. At the end of June of each year t , we use NYSE breakpoints to sort all stocks into deciles based on Accruals for the fiscal year ending in calendar year $t-1$, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$.

NSI (Fama and French, 2008) We measure net stock issues (NSI) as the natural log of the ratio of the split-adjusted shares outstanding scaled by lagged split-adjusted shares outstanding. The split-adjusted shares outstanding is shares outstanding (Compustat annual item CSHO) times the adjustment factor (item AJEX). At the end of June of each year t , we use NYSE breakpoints to sort all stocks into deciles based on NSI for the fiscal year ending in calendar year $t-1$, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$.

$\Delta PI/A$ (Lyandres, Sun, and Zhang, 2008) We measure $\Delta PI/A$ as changes in gross property, plant, and equipment (Compustat annual item PPEGT) plus changes in inventory (item INVT) scaled by lagged total assets (item AT). At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on $\Delta PI/A$ for the fiscal year ending in calendar year $t-1$, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$.

GP/A (Novy-Marx, 2013) We measure GP/A as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS) divided by current total assets (item AT). At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on GP/A for the fiscal year ending in calendar year $t-1$, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$.

ROA (Balakrishnan, Bartov, and Faurel, 2010; Hou, Xue, and Zhang, 2020) We measure ROA as income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged total assets (item ATQ). At the beginning of each month t , we use NYSE breakpoints to sort all stocks into deciles based on ROA computed with the most recently announced quarterly earnings. Monthly value-weighted decile returns are calculated for month t , and the deciles are rebalanced at the beginning of $t+1$. For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to the most recently announced earnings to be within six months prior to the portfolio formation to exclude stale earnings information.

RD/M (Chan, Lakonishok, and Sougiannis, 2001; Hou, Xue, and Zhang, 2020) We measure RD/M as R&D expenses (Compustat annual item XRD) divided by market equity. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on RD/M, which is R&D expenses for the fiscal year ending in calendar year $t-1$ divided by the market equity at the end of December of $t-1$, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$. We keep only firms with positive R&D expenses. Because the accounting treatment of R&D expenses was standardized in 1975, the RD/M decile returns start in July 1976.

Ad/M (Chan, Lakonishok, and Sougiannis, 2001; Hou, Xue, and Zhang, 2020) We measure Ad/M as advertising expenses (Compustat annual item XAD) divided by market equity. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on Ad/M, which is advertising expenses for the fiscal year ending in calendar year $t-1$ divided by the market equity at the end of December of $t-1$, and calculate monthly value-weighted decile returns from

July of year t to June of $t+1$. We keep only firms with positive advertising expenses. Because sufficient XAD data start in 1972, the Ad/M decile returns start in July 1973.

OCA and IOCA (Eisfeldt and Papanikolaou, 2013; Hou, Xue, and Zhang, 2020) We measure OCA and IOCA as organizational capital-to-assets ratio and industry-adjusted organizational capital-to-assets ratio. We construct the stock of organization capital using the perpetual inventory method:

$$OC_{it} = (1 - \delta)OC_{it-1} + SG\&A_{it}/CPI_t, \quad (A.1)$$

where OC_{it} is the organization capital of firm i at the end of year t , $SG\&A_{it}$ is selling, general, and administrative ($SG\&A$) expenses (Compustat annual item XSGA), CPI_t denotes the consumer price index, and δ is the annual depreciation rate of OC . The initial stock of OC is defined as:

$$OC_{i0} = SG\&A_{i0}/(g + \delta), \quad (A.2)$$

where $SG\&A_{i0}$ is the first valid $SG\&A$ observation (zero or positive) for firm i . g is the long-term growth rate of $SG\&A$ and is assumed to be 10% for $SG\&A$. δ is the depreciation rate for OC and is assumed to be 15%. Missing $SG\&A$ values after the starting date are treated as zero. OCA is defined as OC scaled by total assets. To calculate IOCA, we demean a firm's OCA by its industry mean and then divide the demeaned OCA by the standard deviation of OCA within its industry. We use the Fama and French (1997) 17-industry classification. We winsorize OCA at the 1% and 99% levels of all firms each year before the industry standardization to alleviate the impact of outliers. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on OCA and IOCA for the fiscal year ending in calendar year $t-1$, and calculate monthly value-weighted decile returns from July of year t to June of $t+1$. We require $SG\&A$ to be nonmissing in calendar year $t-1$ because this $SG\&A$ receives the highest weight in OC . We also exclude firms with zero OC .

Table 1: Descriptive Properties of Decile Portfolios

This table reports the monthly average excess returns of decile portfolios for 12 anomaly variables, including book-to-market equity ratio (BM), momentum (R11), asset growth (I/A), return-on-equity (ROE), size (Size), accruals (Accruals), net share issues (NSI), investment-to-assets ratio ($\Delta\text{PI}/A$), gross profitability (GP/A), return-on-assets (ROA), R&D-to-market ratio (RD/M), and advertising-to-market ratio (Ad/M). The t -statistics adjusted for heteroscedasticity and autocorrelations are reported in parentheses. Decile portfolios are formed with NYSE breakpoints and value-weighted returns. L denotes the low decile, H the high decile, and H-L the high-minus-low decile. The sample period is from January 1967 to June 2017 for all anomaly variables except for ROA, RD/M, and Ad/M. The sample starts at July 1972, July 1976, and July 1973 for ROA, RD/M, and Ad/M, respectively, due to data availability.

	L	2	3	4	5	6	7	8	9	H	H-L
BM	0.40 (1.83)	0.53 (2.67)	0.60 (3.14)	0.51 (2.48)	0.49 (2.64)	0.54 (2.92)	0.65 (3.41)	0.63 (3.33)	0.72 (3.58)	0.90 (3.77)	0.50 (2.39)
R11	-0.01 (-0.04)	0.34 (1.25)	0.52 (2.40)	0.48 (2.39)	0.48 (2.54)	0.46 (2.38)	0.49 (2.84)	0.60 (3.30)	0.64 (3.11)	1.04 (3.91)	1.05 (3.51)
I/A	0.71 (2.97)	0.70 (3.64)	0.63 (3.74)	0.50 (2.92)	0.55 (3.19)	0.54 (2.96)	0.56 (3.06)	0.50 (2.52)	0.54 (2.27)	0.26 (1.02)	-0.45 (-3.06)
ROE	0.04 (0.12)	0.27 (1.08)	0.42 (1.95)	0.42 (2.30)	0.55 (3.04)	0.48 (2.47)	0.53 (2.76)	0.49 (2.55)	0.55 (2.88)	0.68 (3.28)	0.64 (3.10)
Size	0.84 (2.63)	0.70 (2.49)	0.65 (2.50)	0.64 (2.61)	0.69 (2.83)	0.61 (2.72)	0.61 (2.69)	0.62 (2.96)	0.53 (2.75)	0.46 (2.66)	-0.37 (-1.55)
Accruals	0.58 (2.31)	0.51 (2.48)	0.57 (3.28)	0.47 (2.62)	0.58 (2.96)	0.58 (3.06)	0.59 (2.96)	0.44 (2.09)	0.33 (1.39)	0.19 (0.69)	-0.40 (-2.61)
NSI	0.80 (4.35)	0.61 (3.40)	0.53 (2.77)	0.56 (3.00)	0.50 (2.61)	0.51 (2.63)	0.62 (3.00)	0.58 (2.63)	0.30 (1.33)	0.19 (0.78)	-0.62 (-4.07)
$\Delta\text{PI}/A$	0.75 (3.25)	0.65 (3.37)	0.59 (3.40)	0.54 (2.98)	0.48 (2.53)	0.53 (2.95)	0.53 (2.70)	0.54 (2.61)	0.33 (1.47)	0.34 (1.31)	-0.41 (-2.96)
GP/A	0.28 (1.41)	0.39 (2.05)	0.41 (2.02)	0.42 (2.04)	0.63 (3.18)	0.55 (2.78)	0.47 (2.22)	0.52 (2.46)	0.61 (3.02)	0.63 (3.15)	0.35 (2.36)
ROA	0.01 (0.03)	0.23 (0.83)	0.49 (2.07)	0.45 (2.22)	0.38 (1.85)	0.59 (2.78)	0.52 (2.57)	0.51 (2.47)	0.57 (2.71)	0.59 (2.67)	0.58 (2.73)
RD/M	0.37 (1.85)	0.56 (2.53)	0.45 (1.98)	0.62 (2.72)	0.61 (2.64)	0.66 (3.09)	0.71 (3.16)	0.88 (3.62)	0.74 (2.95)	1.02 (3.11)	0.65 (2.41)
Ad/M	0.42 (1.62)	0.56 (2.32)	0.51 (2.18)	0.64 (2.84)	0.54 (2.55)	0.80 (3.86)	0.64 (3.02)	0.71 (3.14)	0.94 (3.58)	0.91 (3.23)	0.48 (2.08)

Table 2: Descriptive Statistics of Firm-level Accounting Variables in the Fundamental Returns

This table reports the time series averages of the cross-sectional summary statistics (Panel A), including mean, standard deviation, percentiles (5th, 25th, 50th, 75th, and 95th), and pairwise correlations (Panel B) for firm-level annual accounting variables. The sample period of fundamental returns is from June 1967 to December 2016.

Panel A. Summary Statistics							
	Mean	StdDev	p5	p25	p50	p75	p95
I_{it}/K_{it}	0.37	0.48	-0.04	0.11	0.22	0.43	1.43
$\Delta W_{it}/W_{it}$	0.13	0.36	-0.34	-0.06	0.07	0.23	0.88
Y_{it}/K_{it}	7.82	8.72	0.44	2.23	4.91	9.42	31.94
Y_{it}/W_{it}	3.09	1.79	0.70	1.86	2.72	3.95	7.47
$Y_{it}/(K_{it} + W_{it})$	1.64	0.94	0.30	0.97	1.53	2.15	3.78
$K_{it}/(K_{it} + W_{it})$	0.39	0.25	0.06	0.19	0.34	0.57	0.88
w_{it}^B	0.29	0.23	0.01	0.09	0.24	0.45	0.74
δ_{it+1}	0.19	0.13	0.05	0.10	0.15	0.24	0.53
r_{it+1}^B	0.10	0.06	0.02	0.06	0.09	0.12	0.29
Panel B. Pairwise Correlations							
	$\frac{I_{it+1}}{K_{it+1}}$	$\frac{\Delta W_{it+1}}{W_{it+1}}$	$\frac{Y_{it+1}}{K_{it+1}}$	$\frac{Y_{it+1}}{W_{it+1}}$	$\frac{Y_{it+1}}{(K_{it+1} + W_{it+1})}$	w_{it}^B	δ_{it+1}
I_{it}/K_{it}	0.32	0.32	0.16	-0.07	-0.17	-0.18	0.30
I_{it+1}/K_{it+1}		0.24	0.37	0.02	-0.27	-0.27	0.54
$\Delta W_{it}/W_{it}$			0.07	-0.05	-0.07	-0.09	0.06
$\Delta W_{it+1}/W_{it+1}$			0.09	0.27	0.08	-0.13	0.08
Y_{it+1}/K_{it+1}				0.07	-0.65	-0.16	0.17
Y_{it+1}/W_{it+1}					0.47	-0.18	0.10
$Y_{it+1}/(K_{it+1} + W_{it+1})$					-0.35	-0.19	0.05
$K_{it+1}/(K_{it+1} + W_{it+1})$						-0.07	0.21
w_{it}^B						0.32	-0.56
δ_{it+1}							-0.32
							-0.11
							0.13

Table 3: **Parameter Estimates**

Column γ reports the posterior means of the marginal product parameter γ ; column CI_γ reports the 95% credible intervals of γ ; and column $\sigma(\gamma)$ in Panels C and D report the time-series standard deviation of the posterior means of γ when the parameters are time-varying. Similar definitions apply to columns a , CI_a , and $\sigma(a)$ for the adjustment costs parameter a . Column q^K reports the value-weighted average of model-implied marginal q for physical capital, defined as $q_{it}^K \equiv 1 + a_{jt}(1 - \tau_t)I_{it}/K_{it}$ for firm i in industry j at time t . For specifications with time-varying parameters, we report the time-series averages of these statistics due to space constraint.

Industry	γ	CI_γ	$\sigma(\gamma)$	a	CI_a	$\sigma(a)$	q^K
Panel A: Constant parameters θ							
All industries	0.15	[0.15, 0.15]		0.14	[0.13, 0.14]		1.02
Panel B: Industry-specific parameters θ_j							
Consumer Nondurables	0.13	[0.13, 0.13]		0.07	[0.05, 0.09]		1.01
Consumer Durables	0.16	[0.15, 0.16]		0.29	[0.25, 0.34]		1.06
Manufacturing	0.16	[0.16, 0.16]		0.13	[0.11, 0.14]		1.02
Energy	0.20	[0.20, 0.21]		0.10	[0.09, 0.12]		1.01
Business Equipment	0.23	[0.23, 0.24]		0.78	[0.75, 0.81]		1.23
Telecom	0.24	[0.24, 0.25]		0.07	[0.06, 0.08]		1.01
Wholesale & Retail	0.09	[0.08, 0.09]		0.18	[0.16, 0.20]		1.03
Healthcare	0.17	[0.17, 0.18]		0.12	[0.10, 0.15]		1.02
Utilities	0.29	[0.29, 0.30]		0.01	[0.00, 0.02]		1.00
Others	0.15	[0.15, 0.15]		0.15	[0.14, 0.16]		1.03
Panel C: Time-series average of time-varying parameters θ_t							
All industries	0.15	[0.14, 0.15]	0.09	0.29	[0.26, 0.32]	0.30	1.02
Panel D: Time-series average of industry-specific and time-varying parameters θ_{jt}							
Consumer Nondurables	0.13	[0.11, 0.14]	0.09	0.42	[0.29, 0.55]	0.43	1.05
Consumer Durables	0.17	[0.14, 0.19]	0.18	1.15	[0.83, 1.47]	1.13	1.21
Manufacturing	0.16	[0.15, 0.17]	0.11	0.57	[0.51, 0.65]	0.98	1.08
Energy	0.20	[0.18, 0.22]	0.13	0.45	[0.40, 0.48]	0.54	1.06
Business Equipment	0.23	[0.21, 0.24]	0.19	1.78	[1.66, 1.83]	2.00	1.53
Telecom	0.28	[0.25, 0.30]	0.21	0.71	[0.65, 0.76]	0.66	1.09
Wholesale & Retail	0.08	[0.07, 0.09]	0.06	0.87	[0.77, 0.96]	0.98	1.13
Healthcare	0.19	[0.17, 0.21]	0.16	0.59	[0.44, 0.73]	0.68	1.09
Utilities	0.29	[0.25, 0.32]	0.17	0.25	[0.20, 0.32]	0.32	1.02
Others	0.17	[0.15, 0.18]	0.13	0.48	[0.47, 0.54]	0.52	1.09

Table 4: **Summary Statistics of the Realized and Fundamental Firm-level Stock Returns**

This table reports the following key statistics for the realized (r^S) and fundamental (r^F) firm-level stock returns: mean, standard deviation, skewness, kurtosis, the time series average of cross-sectional correlations between the realized and fundamental returns, and mean absolute error (m.a.e.) of the fundamental returns. The m.a.e. is defined as $\text{m.a.e.} \equiv \frac{1}{T} \sum_{t=0}^{T-1} \frac{1}{N_{t+1}} \sum_{i=1}^{N_{t+1}} |r_{it}^S - r_{it}^F|$, where N_{t+1} is the number of firms in period $t + 1$. For fundamental returns, both the posterior means and the 95% credible intervals (in square brackets) of these statistics are reported. Both realized and fundamental returns are winsorized at 0.5 and 99.5 percentiles. The fundamental stock returns are computed based on four model setups: the setup (θ) in which the estimated parameters are constant over time and across industries; the setup (θ_j) in which the estimated parameters are industry-specific but constant over time; the setup (θ_t) in which the estimated parameters are time-varying but constant across industries, and the setup (θ_{jt}) in which the estimated parameters are industry-specific and time-varying. The sample period is from June 1967 to December 2016.

	Data	θ	θ_j	θ_t	θ_{jt}
Mean	14.45	14.89 [14.79, 14.99]	15.47 [15.36, 15.57]	14.97 [14.87, 15.06]	15.65 [15.55, 15.75]
StdDev	60.78	19.76 [19.67, 19.85]	18.49 [18.39, 18.59]	27.36 [27.26, 27.46]	34.17 [34.06, 34.27]
Skewness	2.15	2.12 [2.11, 2.14]	1.68 [1.66, 1.70]	1.66 [1.64, 1.67]	1.68 [1.67, 1.70]
Kurtosis	11.05	13.33 [13.26, 13.41]	10.66 [10.59, 10.73]	10.74 [10.66, 10.82]	11.20 [11.11, 11.29]
Correlation	<i>na</i>	0.09 [0.09, 0.10]	0.12 [0.12, 0.12]	0.12 [0.12, 0.12]	0.20 [0.20, 0.20]
m.a.e	<i>na</i>	42.45 [42.42, 42.48]	41.85 [41.82, 41.89]	40.85 [40.82, 40.88]	40.10 [40.06, 40.13]

Table 5: **Fundamental Anomaly Premiums**

This table reports the annualized fundamental return, r^F , and alpha, defined as the average difference between portfolio realized stock returns and constructed fundamental returns, i.e. $\alpha \equiv r^S - r^F$, of the high minus low hedging portfolios for each of the 12 anomalies and for the four model setups: the setup (θ) in which estimated model parameters are constant over time and across industries; the setup (θ_j) in which estimated model parameters are industry-specific but constant over time; the setup (θ_t) in which estimated model parameters are time-varying but constant across industries; the setup (θ_{jt}) in which estimated model parameters are industry-specific and time-varying. The sample period is from June 1967 to December 2016 for all anomaly variables except for ROA , RD/M , and Ad/M . The sample starts at December 1972, December 1976, and December 1973 for ROA , RD/M , and Ad/M , respectively, due to data availability. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	θ		θ_j		θ_t		θ_{jt}	
	r^F	α	r^F	α	r^F	α	r^F	α
<i>BM</i>	-3.25*	9.99***	-0.64	7.38***	-3.78**	10.52***	0.47	6.27***
	(-1.95)	(4.23)	(-0.52)	(3.25)	(-2.17)	(4.20)	(0.27)	(3.33)
<i>R11</i>	3.82***	9.92***	3.59***	10.16***	5.75***	8.00**	11.88***	1.87
	(4.45)	(2.87)	(5.23)	(3.09)	(5.09)	(2.16)	(11.03)	(0.57)
<i>I/A</i>	0.48	-6.78***	-0.06	-6.24***	-0.68	-5.62***	-3.14**	-3.16**
	(0.48)	(-4.03)	(-0.09)	(-3.42)	(-0.43)	(-3.18)	(-2.29)	(-2.06)
<i>ROE</i>	4.72***	2.97	3.89***	3.80	5.32***	2.36	4.64***	3.05
	(4.88)	(1.11)	(5.28)	(1.46)	(5.90)	(0.90)	(4.26)	(1.33)
<i>Size</i>	-8.10***	3.25	-5.82***	0.98	-7.57***	2.73	-5.98***	1.14
	(-8.84)	(0.93)	(-8.62)	(0.28)	(-6.05)	(0.73)	(-5.62)	(0.34)
<i>Accruals</i>	9.01***	-14.59***	4.90***	-10.48***	8.80***	-14.38***	4.75***	-10.33***
	(16.76)	(-8.08)	(7.96)	(-5.96)	(14.55)	(-7.68)	(4.46)	(-6.29)
<i>NSI</i>	-4.70***	-2.96	-2.80***	-4.85**	-4.59***	-3.07	-3.09***	-4.56***
	(-7.18)	(-1.51)	(-4.28)	(-2.44)	(-6.71)	(-1.49)	(-3.41)	(-2.90)
$\Delta PI/A$	-2.29**	-3.50*	-4.21***	-1.58	-3.12**	-2.67	-5.83***	0.04
	(-2.26)	(-1.90)	(-6.99)	(-0.87)	(-2.07)	(-1.48)	(-4.82)	(0.03)
<i>GP/A</i>	15.28***	-11.42***	6.77***	-2.90	15.48***	-11.61***	7.26***	-3.39***
	(19.27)	(-5.20)	(13.05)	(-1.47)	(14.63)	(-5.41)	(5.84)	(-2.64)
<i>ROA</i>	3.84***	2.62	3.06***	3.40	4.10***	2.35	3.83***	2.63
	(3.66)	(0.95)	(3.40)	(1.27)	(3.86)	(0.89)	(2.84)	(1.11)
<i>RD/M</i>	0.52	8.18**	2.52*	6.18	0.47	8.22**	5.23**	3.47
	(0.49)	(1.97)	(1.92)	(1.49)	(0.50)	(2.07)	(2.12)	(1.42)
<i>Ad/M</i>	14.14***	-8.04***	6.39***	-0.29	13.89***	-7.79**	7.49***	-1.39
	(8.96)	(-2.71)	(5.75)	(-0.10)	(7.32)	(-2.42)	(2.83)	(-0.59)

Table 6: Correlation between Realized Stock Returns and Fundamental Returns

This table reports the contemporaneous correlations between the realized and fundamental returns on the decile portfolios and high-minus-low decile portfolios constructed from the 12 anomaly variables under four specifications. The sample period is from June 1967 to December 2016 for all anomaly variables except for ROA, RD/M, and Ad/M. The sample starts at December 1972, December 1976, and December 1973 for ROA, RD/M, and Ad/M, respectively, due to data availability. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A. Industry-specific and Time-varying Parameters												
	L	2	3	4	5	6	7	8	9	H	H – L	
<i>BM</i>	0.78 ***	0.77 ***	0.74 ***	0.73 ***	0.72 ***	0.69 ***	0.64 ***	0.68 ***	0.63 ***	0.63 ***	0.53 ***	
<i>R11</i>	0.45 ***	0.48 ***	0.63 ***	0.67 ***	0.63 ***	0.67 ***	0.68 ***	0.72 ***	0.67 ***	0.70 ***	0.23 ***	
<i>I/A</i>	0.67 ***	0.73 ***	0.72 ***	0.62 ***	0.73 ***	0.73 ***	0.76 ***	0.71 ***	0.73 ***	0.67 ***	0.49 ***	
<i>ROE</i>	0.59 ***	0.59 ***	0.63 ***	0.65 ***	0.63 ***	0.60 ***	0.67 ***	0.73 ***	0.74 ***	0.71 ***	0.28 ***	
<i>Size</i>	0.57 ***	0.64 ***	0.65 ***	0.68 ***	0.69 ***	0.71 ***	0.71 ***	0.71 ***	0.71 ***	0.77 ***	0.36 ***	
<i>Accruals</i>	0.71 ***	0.73 ***	0.70 ***	0.72 ***	0.71 ***	0.68 ***	0.78 ***	0.77 ***	0.69 ***	0.68 ***	0.41 ***	
<i>NSI</i>	0.69 ***	0.71 ***	0.73 ***	0.72 ***	0.69 ***	0.67 ***	0.74 ***	0.71 ***	0.76 ***	0.67 ***	0.39 ***	
$\Delta PI/A$	0.68 ***	0.72 ***	0.73 ***	0.71 ***	0.73 ***	0.74 ***	0.74 ***	0.77 ***	0.72 ***	0.64 ***	0.52 ***	
<i>GP/A</i>	0.64 ***	0.71 ***	0.72 ***	0.67 ***	0.70 ***	0.72 ***	0.73 ***	0.75 ***	0.80 ***	0.76 ***	0.56 ***	
<i>ROA</i>	0.62 ***	0.65 ***	0.62 ***	0.62 ***	0.64 ***	0.72 ***	0.70 ***	0.70 ***	0.75 ***	0.79 ***	0.27 ***	
<i>RD/M</i>	0.72 ***	0.80 ***	0.71 ***	0.68 ***	0.75 ***	0.67 ***	0.72 ***	0.68 ***	0.68 ***	0.62 ***	0.57 ***	
<i>AD/M</i>	0.72 ***	0.69 ***	0.65 ***	0.67 ***	0.66 ***	0.72 ***	0.66 ***	0.71 ***	0.68 ***	0.58 ***	0.56 ***	
Panel B. Industry-specific Parameters												
	L	2	3	4	5	6	7	8	9	H	H – L	
<i>BM</i>	0.12 ***	0.06	0.06	0.09 **	0.16 ***	0.07	0.07	0.08 *	0.15 ***	0.19 ***	0.36 ***	
<i>R11</i>	0.08 *	-0.06	-0.06	-0.04	-0.02	0.07 *	-0.02	0.10 **	0.04	0.30 ***	0.13 ***	
<i>I/A</i>	0.19 ***	0.12 ***	0.00	-0.09 **	-0.02	0.00	0.01	0.13 ***	0.30 ***	0.12 ***	0.31 ***	
<i>ROE</i>	0.19 ***	0.05	0.02	0.16 ***	-0.05	-0.06	0.08 *	0.05	0.07 *	0.00	0.08 *	
<i>Size</i>	0.15 ***	0.10 **	0.14 ***	-0.01	0.17 ***	0.02	0.08 *	0.12 **	0.08 *	0.03	0.20 ***	
<i>Accruals</i>	0.17 ***	0.30 ***	0.02	0.07 *	-0.07	0.11 ***	-0.07 *	-0.06	0.38 ***	0.20 ***	0.25 ***	
<i>NSI</i>	-0.10 **	0.12 ***	0.00	-0.13 ***	0.07 *	0.17 ***	0.16 ***	0.01	0.21 ***	0.13 ***	-0.02	
$\Delta PI/A$	0.12 ***	-0.07	-0.11 ***	-0.07 *	-0.02	-0.02	0.07 *	0.19 ***	0.08 *	0.11 ***	0.27 ***	
<i>GP/A</i>	-0.03	0.18 **	0.19 ***	0.13 ***	0.14 ***	-0.07 *	0.15 ***	0.00	0.06	0.12 ***	0.06	
<i>ROA</i>	0.09 **	-0.02	0.05	-0.03	-0.09 **	-0.03	-0.06	-0.07	-0.08 *	0.08 *	0.06	
<i>RD/M</i>	-0.03	0.28 ***	-0.15 ***	0.18 ***	0.24 ***	-0.08 *	0.04	-0.20 ***	0.08 *	0.10 **	-0.04	
<i>AD/M</i>	0.28 ***	0.06	-0.05	0.00	-0.08 *	-0.02	-0.08 *	-0.17 **	0.04	0.10 **	0.46 ***	

Panel C. Time-varying Parameters

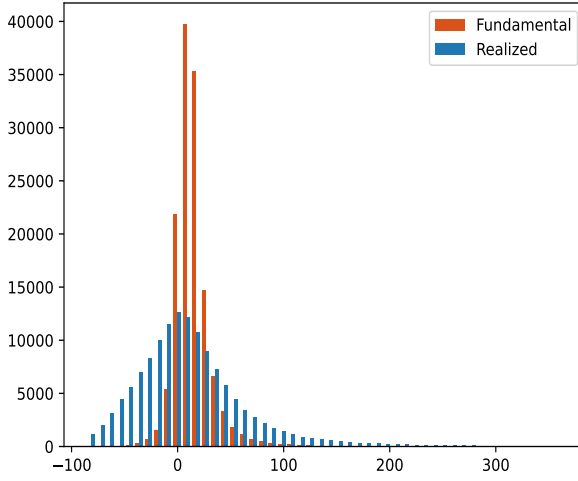
	L	2	3	4	5	6	7	8	9	H	H - L
<i>BM</i>	0.73**	0.78**	0.72**	0.70**	0.66**	0.66**	0.59**	0.58**	0.61**	0.56**	0.24**
<i>R11</i>	0.38**	0.46**	0.58**	0.62**	0.60**	0.68**	0.65**	0.72**	0.66**	0.64**	-0.05
<i>I/A</i>	0.66**	0.71**	0.68**	0.66**	0.72**	0.72**	0.75**	0.72**	0.73**	0.67**	0.37**
<i>ROE</i>	0.56**	0.59**	0.65**	0.65**	0.61**	0.61**	0.71**	0.73**	0.71**	0.70**	0.08
<i>Size</i>	0.55**	0.63**	0.65**	0.67**	0.65**	0.67**	0.70**	0.69**	0.70**	0.74**	0.20**
<i>Accruals</i>	0.65**	0.65**	0.65**	0.69**	0.68**	0.67**	0.74**	0.75**	0.69**	0.67**	0.05
<i>NSI</i>	0.67**	0.71**	0.70**	0.71**	0.68**	0.70**	0.72**	0.68**	0.70**	0.65**	-0.05
$\Delta PI/A$	0.66**	0.75**	0.67**	0.76**	0.65**	0.69**	0.67**	0.74**	0.67**	0.64**	0.33**
<i>GP/A</i>	0.56**	0.61**	0.67**	0.64**	0.68**	0.73**	0.73**	0.74**	0.77**	0.67**	0.13**
<i>ROA</i>	0.58**	0.59**	0.60**	0.56**	0.58**	0.68**	0.67**	0.69**	0.75**	0.76**	0.14**
<i>RD/M</i>	0.66**	0.70**	0.61**	0.65**	0.71**	0.63**	0.68**	0.64**	0.69**	0.45**	-0.02
<i>AD/M</i>	0.68**	0.71**	0.62**	0.61**	0.63**	0.67**	0.48**	0.54**	0.60**	0.53**	0.19**

Panel D. Constant Parameters

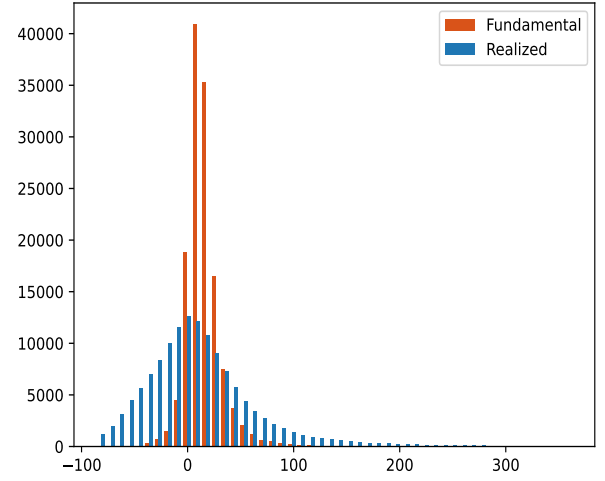
	L	2	3	4	5	6	7	8	9	H	H - L
<i>BM</i>	0.14**	0.04	0.09**	0.06	0.13**	0.18**	0.07*	0.08*	0.13**	0.14**	0.36**
<i>R11</i>	0.00	-0.14**	-0.11**	-0.10**	-0.05	0.02	0.02	0.04	0.03	0.21**	0.01
<i>I/A</i>	0.28**	0.08*	-0.08*	-0.06	0.07	-0.12**	0.08**	0.12**	0.23**	0.14**	0.41**
<i>ROE</i>	0.08*	0.03	0.08**	0.09**	-0.11**	-0.16**	0.04	-0.01	-0.03	0.13**	0.02
<i>Size</i>	0.14**	0.12**	0.09**	-0.04	0.14**	0.02	0.00	0.13**	0.04	0.01	0.18**
<i>Accruals</i>	0.25**	0.19**	0.07	-0.02	-0.07	0.05	-0.04	-0.09**	0.24**	0.11**	0.07
<i>NSI</i>	-0.07*	0.20**	0.06	-0.19**	0.08*	0.16**	0.07*	-0.15**	0.11**	0.05	-0.03
$\Delta PI/A$	0.28**	-0.03	-0.13**	0.00	-0.14**	0.01	0.00	0.20**	0.02	0.09**	0.27**
<i>GP/A</i>	-0.04	0.11**	0.12**	0.13**	0.13**	-0.11**	0.17**	0.00	-0.12**	0.11**	-0.07
<i>ROA</i>	0.00	-0.04	0.12**	-0.13**	-0.14**	-0.11**	-0.19**	-0.08*	-0.11**	0.09**	0.03
<i>RD/M</i>	-0.01	0.18**	-0.11**	0.08*	0.22**	-0.10**	-0.04	-0.21**	0.11**	-0.07	-0.12**
<i>AD/M</i>	0.16**	0.10**	0.01	-0.16**	-0.06	-0.03	-0.14**	-0.20**	0.08*	0.10**	0.36**

Figure 1: **Distribution of Firm-level Returns: Realized vs. Fundamental**

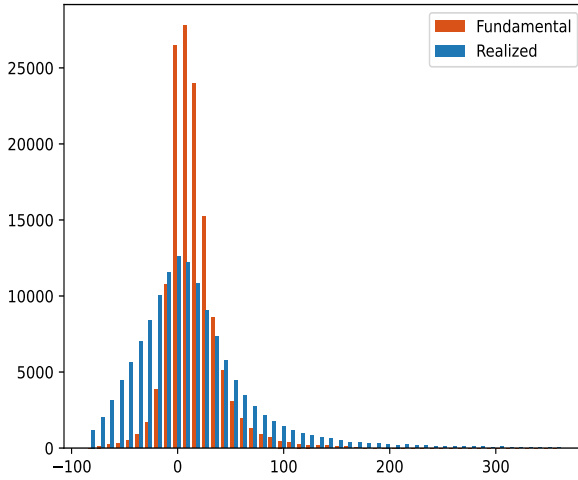
This table presents the histograms of the realized (in blue) and fundamental (in orange) firm-level stock returns from June 1967 to December 2016 under the specifications with constant, industry-specific, time-varying, and industry-specific and time-varying parameter values in Panel (a) - (d), respectively. Returns are in percentage per annum. The number of observations is 136,598. Observations are trimmed at 0.5 and 99.5 percentiles for illustration purposes.



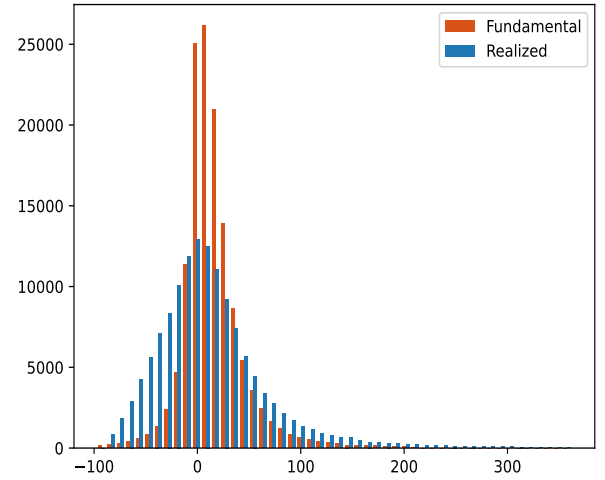
(a) Constant parameters



(b) Industry-specific parameters



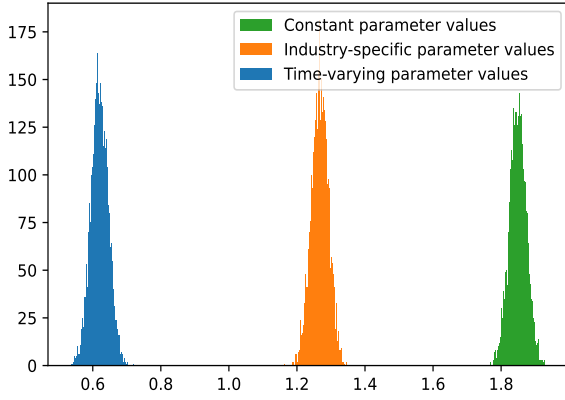
(c) Time-varying parameter values



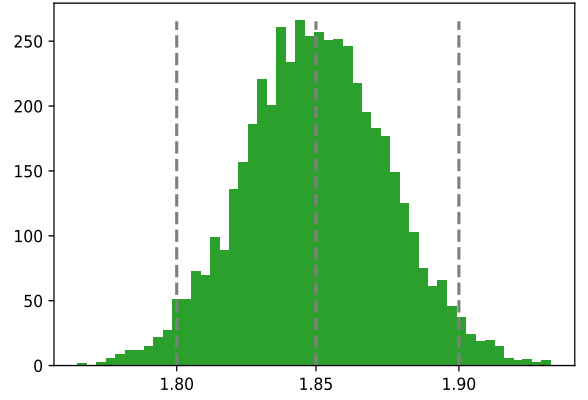
(d) Industry-specific and time-varying parameters

Figure 2: **Performance Comparison**

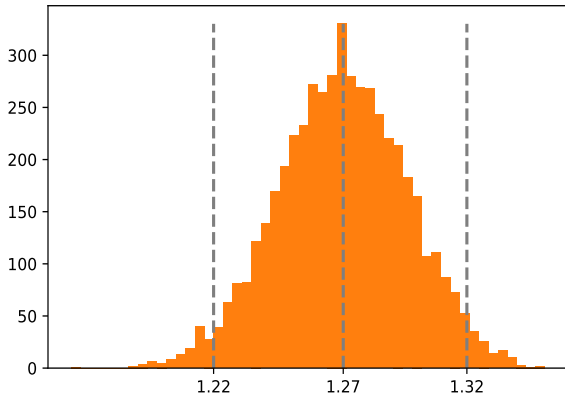
This figure plots the posterior distributions of the differences in m.a.e. among the four specifications. The differences in m.a.e. between the benchmark specification, defined as the one with the lowest m.a.e. (i.e., the specification with industry-specific and time-varying parameters) and an alternative specification is given by $d^a = \frac{1}{T} \sum_{t=1}^T \frac{1}{N_t} \sum_{i=1}^{N_t} \left(\left| r_{it+1}^S - r_{it+1}^{F(a)} \right| - \left| r_{it+1}^S - r_{it+1}^{F(b)} \right| \right)$, where $r^{F(b)}$ and $r^{F(a)}$ are the fundamental returns under the benchmark specification b and under an alternative specification a , respectively, and d^a is in percentage per annum. The three alternative specifications are the ones with constant, industry-specific, and time-varying parameter values, respectively. The black dash lines in panels (b)-(d) mark the 2.5, 50, and 97.5 percentiles of the posterior distributions of the corresponding d^a .



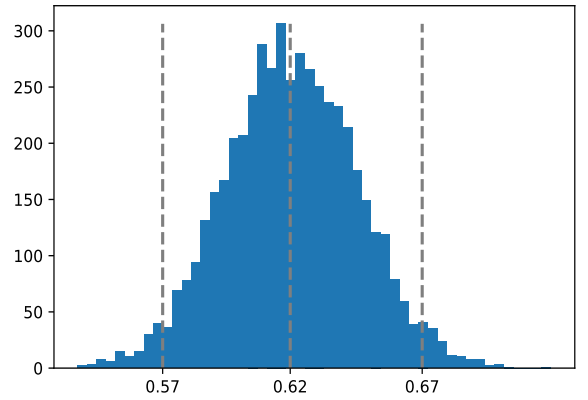
(a) Three alternative estimations



(b) Constant parameters



(c) Industry-specific parameters



(d) Time-varying parameters

Figure 3: **Posterior Distributions of the Fundamental Factor Premiums**

This figure plots the posterior probability density functions of the fundamental factor premiums formed on book-to-market (BM), momentum (R11), asset growth (I/A), return-on-equity (ROE), size (Size), accruals (Accruals), net share issues (NSI), investment-to-assets ratio ($\Delta PI/A$), gross profitability (GP/A), return-on-assets (ROA), R&D-to-market ratio (RD/M), and advertising-to-market ratio (Ad/M). The red lines represent normal distributions with means and standard deviations being the posterior means and standard deviations of the corresponding factor premiums. The 2.5%, 50%, and 97.5% percentiles of each posterior distribution are labeled below the horizontal axes.

